

#1 · Nutrition and Preventive Care

At what age does a healthy infant typically regain birth weight after the expected early weight loss?

Answer

By about 2 weeks of age, after losing up to about 10% of birth weight in the first week.

#2 · Nutrition and Preventive Care

By what age should the Moro reflex normally disappear?

Answer

By about 4 to 6 months of age; persistence beyond this suggests upper motor neuron dysfunction.

#3 · Nutrition and Preventive Care

What is the classic first sign of puberty in females, and at what age range does it usually begin?

Answer

Thelarche (breast budding), usually between ages 8 and 13.

#4 · Nutrition and Preventive Care

What is the classic first sign of puberty in males, and at what age range does it usually begin?

Answer

Testicular enlargement, usually between ages 9 and 14.

#5 · Nutrition and Preventive Care

How long after thelarche does menarche typically occur?

Answer

About 2 to 2.5 years after thelarche begins.

#6 · Nutrition and Preventive Care

What defines precocious puberty in girls and in boys?

Answer

Onset of pubertal signs before age 8 in girls or before age 9 in boys.

#7 · Nutrition and Preventive Care

What defines delayed puberty in girls and in boys?

Answer

Absence of any pubertal signs by age 13 in girls or age 14 in boys.

#8 · Nutrition and Preventive Care

What are the two clinical forms of thiamine (vitamin B1) deficiency and what does each affect?

Answer

Wet beriberi affects the heart (high-output failure); dry beriberi affects peripheral nerves (symmetric weakness and sensory loss).

#9 · Nutrition and Preventive Care

What triad defines Wernicke encephalopathy?

Answer

Confusion, ataxia, and ophthalmoplegia (with nystagmus), from thiamine deficiency.

#10 · Nutrition and Preventive Care

What must be given before or with intravenous glucose in a patient at risk for Wernicke encephalopathy?

Answer

Thiamine, since glucose alone can precipitate or worsen the encephalopathy.

#11 · Nutrition and Preventive Care

What is Korsakoff syndrome and how does it relate to Wernicke encephalopathy?

Answer

A chronic, often irreversible amnestic disorder (confabulation, anterograde amnesia) that can follow inadequately treated Wernicke encephalopathy.

#12 · Nutrition and Preventive Care

What is the classic triad of pellagra and which vitamin deficiency causes it?

Answer

Dermatitis (photosensitive rash), diarrhea, and dementia, from niacin (vitamin B3) deficiency.

#13 · Nutrition and Preventive Care

Which two deficiencies cause megaloblastic anemia, and what key lab value distinguishes them?

Answer

Folate and vitamin B12 deficiency; only B12 deficiency also raises methylmalonic acid (both raise homocysteine).

#14 · Nutrition and Preventive Care

What neurologic syndrome is specific to vitamin B12 deficiency (not folate deficiency)?

Answer

Subacute combined degeneration of the spinal cord, affecting dorsal columns and lateral corticospinal tracts.

#15 · Nutrition and Preventive Care

Why is folic acid supplementation recommended before conception?

Answer

To reduce the risk of neural tube defects such as spina bifida in early pregnancy.

#16 · Nutrition and Preventive Care

What are the classic findings of scurvy and which vitamin deficiency causes it?

Answer

Bleeding gums, perifollicular hemorrhages, poor wound healing, and corkscrew hairs, from vitamin C deficiency (impaired collagen cross-linking).

#17 · Nutrition and Preventive Care

What does vitamin D deficiency cause in children versus in adults?

Answer

Rickets in children (bowed legs, rachitic rosary, delayed fontanelle closure); osteomalacia in adults (bone pain, proximal muscle weakness, poor mineralization).

#18 · Nutrition and Preventive Care

Which clotting factors and proteins depend on vitamin K for synthesis?

Answer

Factors 2, 7, 9, and 10, along with proteins C and S.

#19 · Nutrition and Preventive Care

Why do newborns receive a vitamin K injection at birth?

Answer

To prevent hemorrhagic disease of the newborn, since infants have low vitamin K stores and a sterile gut at birth.

#20 · Nutrition and Preventive Care

What does chronic vitamin A toxicity cause?

Answer

Headache, blurred vision (pseudotumor cerebri), alopecia, dry skin, and hepatotoxicity; it is also teratogenic.

#21 · Nutrition and Preventive Care

What does vitamin A deficiency cause?

Answer

Night blindness, xerophthalmia, and Bitot spots, which can progress to corneal ulceration and blindness.

#22 · Nutrition and Preventive Care

What is the classic triad of acrodermatitis enteropathica or acquired zinc deficiency?

Answer

Dermatitis (perioral and acral), diarrhea, and alopecia, often with additional features such as poor wound healing and hypogeusia (impaired taste).

#23 · Nutrition and Preventive Care

What is pica and with which common deficiency is it associated?

Answer

Persistent craving and consumption of non-nutritive substances (such as ice); associated with iron deficiency anemia.

#24 · Nutrition and Preventive Care

What nail finding is classically associated with chronic iron deficiency anemia?

Answer

Koilonychia (spoon-shaped nails).

#25 · Nutrition and Preventive Care

What is the mechanism of pellagra in carcinoid syndrome?

Answer

Excess tryptophan is diverted to serotonin synthesis by the tumor, leaving too little tryptophan for niacin synthesis.

#26 · Nutrition and Preventive Care

What electrolyte disturbances define refeeding syndrome, and what is the underlying mechanism?

Answer

Hypophosphatemia, hypokalemia, and hypomagnesemia, caused by an insulin surge driving these electrolytes intracellularly when nutrition is reintroduced after starvation.

#27 · Nutrition and Preventive Care

How should nutrition be reintroduced in a severely malnourished patient to prevent refeeding syndrome?

Answer

Start feeding slowly with gradual advancement while proactively monitoring and repleting phosphate, potassium, and magnesium.

#28 · Nutrition and Preventive Care

What are the core diagnostic features of anorexia nervosa?

Answer

Significantly low body weight, intense fear of weight gain, and distorted body image, often with amenorrhea and bradycardia.

#29 · Nutrition and Preventive Care

What are the core diagnostic features of bulimia nervosa?

Answer

Recurrent binge eating with compensatory behavior (such as self-induced vomiting), typically at a normal or near-normal body weight.

#30 · Nutrition and Preventive Care

What physical exam finding (Russell sign) is relatively specific to self-induced vomiting?

Answer

Calluses or scarring on the dorsal surface of the fingers used to induce vomiting.

#31 · Nutrition and Preventive Care

What laboratory abnormality is classic in bulimia nervosa from repeated vomiting?

Answer

Hypokalemic metabolic alkalosis.

#32 · Nutrition and Preventive Care

What defines binge eating disorder, and how does it differ from bulimia nervosa?

Answer

Recurrent binge eating episodes without regular compensatory behaviors (unlike bulimia nervosa, which includes purging, laxative misuse, or excessive exercise).

#33 · Nutrition and Preventive Care

What are the body mass index cutoffs for underweight, normal, overweight, and obesity?

Answer

Underweight under 18.5; normal 18.5 to 24.9; overweight 25 to 29.9; obesity 30 or above (class 3, or severe, obesity is 40 or above).

#34 · Nutrition and Preventive Care

What is the first-line management approach for obesity before considering pharmacotherapy or surgery?

Answer

Intensive lifestyle intervention (diet and exercise counseling).

#35 · Nutrition and Preventive Care

What are general body mass index thresholds supporting consideration of bariatric surgery?

Answer

A body mass index of 40 or above alone, or around 35 (or lower with some criteria) with a significant obesity-related comorbidity, after an inadequate response to lifestyle and pharmacologic measures.

#36 · Nutrition and Preventive Care

At what age is average-risk mammography screening generally recommended to begin?

Answer

Somewhere in the range of age 40 to 50, continuing every 1 to 2 years through at least the mid-70s.

#37 · Nutrition and Preventive Care

What is the recommended cervical cancer screening strategy for women age 21 to 29?

Answer

Cytology alone every 3 years.

#38 · Nutrition and Preventive Care

What are the accepted cervical cancer screening strategies for women age 30 to 65?

Answer

Cytology alone every 3 years, high-risk HPV testing alone every 5 years, or co-testing every 5 years.

#39 · Nutrition and Preventive Care

At what age can cervical cancer screening generally stop, and under what condition?

Answer

Around age 65, if there has been adequate prior negative screening and no history of high-grade dysplasia.

#40 · Nutrition and Preventive Care

At what age does average-risk colorectal cancer screening begin, and what are two accepted modalities?

Answer

Age 45; colonoscopy every 10 years or annual fecal immunochemical testing, among other options.

#41 · Nutrition and Preventive Care

Who qualifies for annual low-dose CT lung cancer screening?

Answer

Adults age 50 to 80 with a significant smoking history who currently smoke or quit within the past 15 years.

#42 · Nutrition and Preventive Care

At what age should routine DEXA osteoporosis screening begin in women?

Answer

Age 65 for all women, earlier in postmenopausal women under 65 with risk factors.

#43 · Nutrition and Preventive Care

What T-score defines osteoporosis, osteopenia, and normal bone density?

Answer

Osteoporosis is negative 2.5 or below; osteopenia is negative 1.0 to negative 2.5; normal is above negative 1.0.

#44 · Nutrition and Preventive Care

Who qualifies for one-time abdominal aortic aneurysm screening ultrasound?

Answer

Men age 65 to 75 who have ever smoked.

#45 · Nutrition and Preventive Care

What is the CAGE questionnaire used to screen for, and what does a positive result mean?

Answer

Alcohol use disorder (Cut down, Annoyed, Guilty, Eye-opener); 2 or more yes answers suggests a possible alcohol use disorder.

#46 · Nutrition and Preventive Care

What does the 5 A's framework structure, and what are its steps?

Answer

Brief tobacco cessation counseling: Ask, Advise, Assess, Assist, Arrange.

#47 · Nutrition and Preventive Care

How does combining behavioral counseling with pharmacotherapy affect tobacco quit rates?

Answer

It roughly doubles quit rates compared with either approach used alone.

#48 · Nutrition and Preventive Care

What does a positive PHQ-2 screen prompt as a next step?

Answer

A more detailed depression assessment, such as the PHQ-9.

#49 · Nutrition and Preventive Care

What is the Timed Up and Go test used for, and what result suggests increased fall risk?

Answer

Assessing fall risk in older adults; a time greater than about 12 seconds suggests increased risk.

#50 · Nutrition and Preventive Care

What tool is used for brief cognitive screening at a Medicare Annual Wellness Visit?

Answer

The Mini-Cog or Mini-Mental State Examination.

#51 · Nutrition and Preventive Care

What are the four categories of elder mistreatment?

Answer

Physical abuse, emotional abuse, financial exploitation, and neglect (self-neglect is a related but distinct category).

#52 · Nutrition and Preventive Care

What is the recommended immediate step when elder abuse is suspected but a caregiver dominates the visit?

Answer

Interview the patient alone, without the caregiver present, before proceeding further.

#53 · Nutrition and Preventive Care

What is the DTaP vaccination schedule in infancy and early childhood?

Answer

2, 4, 6, and 15 to 18 months, with a booster at 4 to 6 years, then transitioning to Tdap in adolescence.

#54 · Nutrition and Preventive Care

At what ages are MMR and varicella vaccines routinely given?

Answer

12 to 15 months, with a second dose at 4 to 6 years.

#55 · Nutrition and Preventive Care

Why are MMR, varicella, and rotavirus vaccines all contraindicated in significant immunosuppression?

Answer

They are all live attenuated vaccines, which carry a risk of vaccine-strain disease in an immunosuppressed host.

#56 · Nutrition and Preventive Care

Why does the rotavirus vaccine have a strict maximum age for starting the series?

Answer

Because of an observed association with intussusception when given outside its recommended age window.

#57 · Nutrition and Preventive Care

How many doses of the HPV vaccine are needed if started before age 15, versus at or after age 15?

Answer

2 doses if started before age 15; 3 doses if started at 15 or older, or in immunocompromised patients.

#58 · Nutrition and Preventive Care

At what age is the recombinant zoster (shingles) vaccine recommended to begin?

Answer

Age 50.

#59 · Nutrition and Preventive Care

Which vaccine is recommended during every pregnancy regardless of prior vaccination history?

Answer

Tdap, ideally in the early third trimester, to protect the newborn from pertussis.

#60 · Nutrition and Preventive Care

What tetanus booster interval applies to a clean, minor wound versus a contaminated or high-risk wound?

Answer

10 years for a clean minor wound; 5 years for a contaminated or high-risk wound.

#61 · Nutrition and Preventive Care

Which vaccine is required for entry into certain countries in sub-Saharan Africa and South America?

Answer

Yellow fever vaccine (a live vaccine).

#62 · Nutrition and Preventive Care

What are three commonly used oral malaria prophylaxis regimens?

Answer

Atovaquone-proguanil, doxycycline, and mefloquine, selected based on regional resistance patterns.

#63 · Nutrition and Preventive Care

What vaccine is recommended for most travelers to destinations with unreliable water treatment who are not already immune?

Answer

Hepatitis A vaccine.

#64 · Nutrition and Preventive Care

When is enteral nutrition preferred over parenteral nutrition?

Answer

Whenever the gastrointestinal tract is functional, since it preserves gut mucosal integrity and carries a lower infection risk.

#65 · Nutrition and Preventive Care

When is parenteral nutrition indicated instead of enteral nutrition?

Answer

When the gastrointestinal tract is nonfunctional, such as with bowel obstruction, severe ileus, or short bowel syndrome.

#66 · Nutrition and Preventive Care

What are two major risks of long-term parenteral nutrition?

Answer

Catheter-related bloodstream infection and hepatic steatosis (liver dysfunction).

#67 · Nutrition and Preventive Care

Why is serum albumin an unreliable isolated marker of nutritional status in an acutely ill patient?

Answer

Because it is also a negative acute-phase reactant and falls with inflammation independent of true nutritional status.

#68 · Nutrition and Preventive Care

What waist circumference measurement adds to body mass index in cardiometabolic risk assessment?

Answer

An estimate of visceral (central) adiposity and associated cardiometabolic risk, independent of body mass index alone.

#69 · Nutrition and Preventive Care

At what age should routine diabetes screening generally begin in an average-risk adult without risk factors?

Answer

Around age 35, or earlier with risk factors such as overweight, obesity, or a strong family history.

#70 · Nutrition and Preventive Care

What is the purpose of calculating an atherosclerotic cardiovascular disease risk score?

Answer

To help guide the decision of whether to initiate statin therapy for primary prevention.

#71 · Social Sciences Ethics Law and Patient Safety

What four elements must a patient understand for informed consent to be valid?

Answer

The proposed intervention, its risks and benefits, the reasonable alternatives (including doing nothing), and enough understanding to make a voluntary choice.

#72 · Social Sciences Ethics Law and Patient Safety

What four abilities define decision-making capacity?

Answer

The ability to understand relevant information, appreciate how it applies to one's own situation, reason through the options, and communicate a consistent choice.

#73 · Social Sciences Ethics Law and Patient Safety

Who determines capacity, and who determines competence?

Answer

Any treating physician can assess capacity. Competence is a legal status that only a court can determine.

#74 · Social Sciences Ethics Law and Patient Safety

Is capacity a global, permanent status?

Answer

No. Capacity is decision-specific and time-specific; a patient can have capacity for one decision but not another, or regain capacity during a lucid interval.

#75 · Social Sciences Ethics Law and Patient Safety

What is implied consent?

Answer

The legal doctrine allowing treatment without explicit consent in an emergency, when a patient cannot communicate and delay would cause serious harm.

#76 · Social Sciences Ethics Law and Patient Safety

What is therapeutic privilege?

Answer

A narrow exception allowing a physician to withhold specific information if disclosure would cause serious psychological harm; it does not justify withholding a diagnosis simply because it is upsetting.

#77 · Social Sciences Ethics Law and Patient Safety

What is waiver, in the context of informed consent?

Answer

A patient's choice to decline detailed information and ask that a physician or family member decide on their behalf instead.

#78 · Social Sciences Ethics Law and Patient Safety

Name common exceptions to requiring parental consent for a minor's care.

Answer

Emancipated minors, and care for sexually transmitted infections, contraception and pregnancy, substance abuse treatment, and emergency care.

#79 · Social Sciences Ethics Law and Patient Safety

What is the default legal surrogate hierarchy when a patient lacks capacity and has no guardian or named health care agent?

Answer

Spouse, then adult children, then parents, then adult siblings, then other relatives. A court-appointed guardian or a named health care agent, if one already exists, takes priority over this entire list.

#80 · Social Sciences Ethics Law and Patient Safety

What is assent, and how does it differ from consent for a minor?

Answer

Assent is a child's own agreement and cooperation with a proposed treatment; the parent or guardian still provides the legally binding consent.

#81 · Social Sciences Ethics Law and Patient Safety

What does HIPAA stand for and what does it establish?

Answer

The Health Insurance Portability and Accountability Act; it sets the federal floor for protecting health information, with stricter state laws taking precedence when they exist.

#82 · Social Sciences Ethics Law and Patient Safety

Name conditions requiring mandatory reporting that override confidentiality.

Answer

Suspected child abuse or neglect, suspected elder or vulnerable-adult abuse, and certain communicable diseases reportable to public health authorities.

#83 · Social Sciences Ethics Law and Patient Safety

What is the Tarasoff principle?

Answer

A physician's duty to take reasonable steps to protect an identifiable third party from a patient's credible, specific threat of harm, which may include warning the victim or notifying police.

#84 · Social Sciences Ethics Law and Patient Safety

Is intimate partner violence against a competent adult generally subject to mandatory reporting?

Answer

No, most states do not mandate reporting for a competent adult victim who does not want to report, preserving the adult's autonomy over their own safety decisions.

#85 · Social Sciences Ethics Law and Patient Safety

Can a minor's disclosure of sexual activity generally be shared with parents without permission?

Answer

No, most states protect this disclosure as confidential, one of the recognized exceptions to parental notification.

#86 · Social Sciences Ethics Law and Patient Safety

What is the HIPAA minimum necessary standard?

Answer

Even when disclosure is permitted, only the minimum amount of information needed for that specific purpose should be shared.

#87 · Social Sciences Ethics Law and Patient Safety

Do psychotherapy notes receive extra protection under HIPAA?

Answer

Yes, they generally require a separate, specific authorization before release, beyond what is needed for the rest of the medical record.

#88 · Social Sciences Ethics Law and Patient Safety

What are the three core purposes for which HIPAA allows information to flow without a separate authorization?

Answer

Treatment, payment, and health care operations.

#89 · Social Sciences Ethics Law and Patient Safety

What is a living will?

Answer

An advance directive describing a patient's specific treatment preferences, such as whether they want mechanical ventilation or artificial nutrition in a terminal or permanently unconscious state.

#90 · Social Sciences Ethics Law and Patient Safety

What is a durable power of attorney for health care?

Answer

A document designating a specific person, a health care agent, to make medical decisions on the patient's behalf if the patient loses capacity.

#91 · Social Sciences Ethics Law and Patient Safety

When a living will and a later-named health care agent conflict, whose judgment generally controls?

Answer

The health care agent's judgment about the patient's likely current wishes generally takes priority, since the agent can apply new information the document could not anticipate.

#92 · Social Sciences Ethics Law and Patient Safety

What is the substituted judgment standard?

Answer

A surrogate decides as the patient themselves would have decided, based on the patient's own previously expressed values, not the surrogate's personal preference.

#93 · Social Sciences Ethics Law and Patient Safety

What is the best interest standard, and when is it used?

Answer

A surrogate chooses the option a reasonable person would want in that situation; it is used only when the patient's own wishes are entirely unknown.

#94 · Social Sciences Ethics Law and Patient Safety

Does a DNR order mean a patient should receive no other treatment?

Answer

No. A DNR order addresses only cardiopulmonary resuscitation; all other appropriate care, such as antibiotics or surgery, should continue unless separately declined.

#95 · Social Sciences Ethics Law and Patient Safety

What is medical futility?

Answer

An intervention that cannot achieve its intended physiologic goal or would not meaningfully benefit the patient given their overall prognosis.

#96 · Social Sciences Ethics Law and Patient Safety

What is the appropriate step when a physician and family disagree about withdrawing futile treatment?

Answer

Involve the hospital ethics committee to mediate the disagreement and clarify the medical facts and the patient's own values.

#97 · Social Sciences Ethics Law and Patient Safety

How does physician-assisted death differ from euthanasia?

Answer

In physician-assisted death the patient self-administers the lethal medication; in euthanasia another person administers it directly. Euthanasia is illegal everywhere in the United States.

#98 · Social Sciences Ethics Law and Patient Safety

What is the principle of double effect?

Answer

An action with a good intended effect, such as relieving severe pain with opioids, is ethically permissible even if it carries a foreseen but unintended risk of a harmful secondary effect, as long as causing harm is not the intent.

#99 · Social Sciences Ethics Law and Patient Safety

Is withdrawing a non-beneficial, life-sustaining treatment considered euthanasia?

Answer

No. Withdrawing or withholding a treatment that is not providing meaningful benefit is standard, accepted practice, since the underlying disease, not the withdrawal, causes death.

#100 · Social Sciences Ethics Law and Patient Safety

What is a POLST form?

Answer

Physician Orders for Life-Sustaining Treatment, a portable medical order reflecting a patient's goals of care that should be interpreted in the context of those goals rather than applied rigidly to every clinical situation.

#101 · Social Sciences Ethics Law and Patient Safety

What often automatically revokes a former spouse's authority as a designated health care agent?

Answer

Divorce, in most states, unless the patient specifically reaffirms the designation afterward.

#102 · Social Sciences Ethics Law and Patient Safety

What is the Swiss cheese model of error?

Answer

A model in which multiple imperfect layers of defense each have small gaps, and a serious error usually requires several gaps to align at once to reach the patient.

#103 · Social Sciences Ethics Law and Patient Safety

What is a root cause analysis?

Answer

A structured, retrospective, multidisciplinary review after a serious event that identifies the underlying systemic contributing factors, continuing past the most obvious proximate cause.

#104 · Social Sciences Ethics Law and Patient Safety

What is a PDSA cycle?

Answer

Plan-Do-Study-Act, an iterative quality improvement method: plan a small testable change, implement it, study the results, then decide whether to adopt, adapt, or abandon it.

#105 · Social Sciences Ethics Law and Patient Safety

What is a near miss or close call?

Answer

An error caught before it reaches the patient and causes no harm; it remains valuable safety data and should still be formally reported.

#106 · Social Sciences Ethics Law and Patient Safety

What is required when a medical error reaches a patient?

Answer

Prompt, honest disclosure, including an acknowledgment that an error occurred, an apology, and a description of steps being taken to prevent recurrence, regardless of whether harm resulted.

#107 · Social Sciences Ethics Law and Patient Safety

Does transparent error disclosure increase or decrease malpractice litigation risk?

Answer

Most current evidence suggests transparent disclosure actually reduces malpractice litigation rather than increasing it.

#108 · Social Sciences Ethics Law and Patient Safety

What is a just culture?

Answer

A framework that distinguishes ordinary human error and at-risk behavior, addressed through system improvement, from reckless conduct, which still warrants individual accountability.

#109 · Social Sciences Ethics Law and Patient Safety

What is a sentinel event?

Answer

An unexpected occurrence involving serious harm or death not primarily related to the natural course of the patient's illness, requiring a formal root cause analysis.

#110 · Social Sciences Ethics Law and Patient Safety

What defines a systems-based approach to patient safety?

Answer

Recognizing that most serious errors result from multiple aligned weaknesses in a system's design, rather than from a single individual's isolated failure.

#111 · Social Sciences Ethics Law and Patient Safety

What does Medicare Part A cover?

Answer

Inpatient hospital care, skilled nursing facility stays, hospice, and some home health care, funded through payroll taxes with generally no monthly premium for most beneficiaries.

#112 · Social Sciences Ethics Law and Patient Safety

What does Medicare Part B cover?

Answer

Outpatient physician services, durable medical equipment, and preventive services, requiring a monthly premium.

#113 · Social Sciences Ethics Law and Patient Safety

What is Medicare Part C?

Answer

Medicare Advantage, an alternative to original Medicare in which a private insurer administers the benefit, often bundling in additional coverage.

#114 · Social Sciences Ethics Law and Patient Safety

What does Medicare Part D cover?

Answer

Outpatient prescription drugs, administered through private plans.

#115 · Social Sciences Ethics Law and Patient Safety

What is Medicaid?

Answer

A joint federal and state program providing coverage for eligible low-income individuals and families, with eligibility and covered benefits varying by state.

#116 · Social Sciences Ethics Law and Patient Safety

How does an HMO generally work?

Answer

It requires selecting a primary care physician who coordinates care and provides referrals to specialists, and generally does not cover out-of-network care except in emergencies.

#117 · Social Sciences Ethics Law and Patient Safety

How does a PPO generally work?

Answer

It allows seeing specialists without a referral and provides at least partial coverage for out-of-network care, usually with a higher monthly premium.

#118 · Social Sciences Ethics Law and Patient Safety

What is capitation?

Answer

A reimbursement model paying a fixed amount per patient per period regardless of services used, shifting financial risk to the provider and incentivizing efficient, preventive care.

#119 · Social Sciences Ethics Law and Patient Safety

What is fee-for-service reimbursement?

Answer

A model that pays a separate amount for each individual service rendered, which can incentivize higher volumes of testing and treatment.

#120 · Social Sciences Ethics Law and Patient Safety

What is an accountable care organization?

Answer

A group of providers who share responsibility for the cost and quality of care for a defined population, aligning incentives with outcomes rather than service volume.

#121 · Social Sciences Ethics Law and Patient Safety

What is a conflict of interest, and when must it be disclosed?

Answer

Any situation where personal, financial, or professional interests could improperly influence professional judgment; it must be disclosed simply because it creates the potential for bias, not only if bias is proven.

#122 · Social Sciences Ethics Law and Patient Safety

What is the role of an institutional review board?

Answer

To review proposed human-subjects research before it begins, evaluating the risk-benefit balance, informed consent procedures, and additional protections for vulnerable populations.

#123 · Social Sciences Ethics Law and Patient Safety

What must research informed consent specifically disclose, beyond standard clinical consent?

Answer

That participation is voluntary, that the subject may withdraw at any time without penalty, and any risks specific to the research itself.

#124 · Social Sciences Ethics Law and Patient Safety

Are romantic relationships with current patients ever appropriate?

Answer

No, they are never appropriate and violate the professional boundary inherent in the physician-patient relationship.

#125 · Social Sciences Ethics Law and Patient Safety

What should a physician do upon recognizing signs of impairment in a colleague?

Answer

Report the concern through the appropriate institutional or state physician health channel, rather than ignoring it or handling it only informally.

#126 · Renal Urinary and Reproductive Systems

What are the three categories used to classify acute kidney injury?

Answer

Prerenal (reduced kidney perfusion), intrinsic renal (damage to tubules, interstitium, or glomeruli), and postrenal (obstruction of urine flow).

#127 · Renal Urinary and Reproductive Systems

What FENa value suggests prerenal azotemia versus acute tubular necrosis?

Answer

FENa under 1 percent suggests prerenal azotemia (avid sodium reabsorption). FENa above 2 percent suggests acute tubular necrosis (loss of tubular function).

#128 · Renal Urinary and Reproductive Systems

What urine sediment finding is classic for acute tubular necrosis?

Answer

Muddy brown granular casts, reflecting sloughed tubular epithelial cells.

#129 • Renal Urinary and Reproductive Systems

What triad suggests drug-induced acute interstitial nephritis?

Answer

Fever, rash, and eosinophiluria following a new medication such as a beta-lactam, sulfa drug, PPI, or NSAID.

#130 • Renal Urinary and Reproductive Systems

What does the mnemonic AEIOU stand for in dialysis indications?

Answer

Refractory Acidosis, severe Electrolyte abnormalities (hyperkalemia), Ingestion of a dialyzable toxin, volume Overload unresponsive to diuretics, and Uremic symptoms.

#131 • Renal Urinary and Reproductive Systems

What are the main complications of chronic kidney disease?

Answer

Normocytic anemia (low erythropoietin), secondary hyperparathyroidism and renal osteodystrophy, metabolic acidosis, and uremic symptoms such as pericarditis and platelet dysfunction.

#132 • Renal Urinary and Reproductive Systems

Why are ACE inhibitors and ARBs preferred in proteinuric chronic kidney disease?

Answer

They dilate the efferent arteriole preferentially, lowering intraglomerular pressure and reducing proteinuria beyond their blood pressure lowering effect.

#133 · Renal Urinary and Reproductive Systems

How is nephrotic syndrome defined?

Answer

Proteinuria over 3.5 grams per day, hypoalbuminemia, edema, and hyperlipidemia, with a relatively bland urine sediment.

#134 · Renal Urinary and Reproductive Systems

What is the most common cause of nephrotic syndrome in children, and what does biopsy show?

Answer

Minimal change disease; light microscopy is normal, but electron microscopy shows podocyte foot process effacement. It responds well to corticosteroids.

#135 • Renal Urinary and Reproductive Systems

Which nephrotic syndrome is most associated with heroin use, HIV, and Black patients?

Answer

Focal segmental glomerulosclerosis (FSGS), showing segmental sclerosis of some glomeruli on biopsy.

#136 • Renal Urinary and Reproductive Systems

Which nephrotic syndrome is associated with solid tumors, hepatitis B, and anti-PLA2R antibodies?

Answer

Membranous nephropathy, which also carries an elevated risk of renal vein thrombosis.

#137 · Renal Urinary and Reproductive Systems

What biopsy finding is classic for diabetic nephropathy?

Answer

Nodular glomerulosclerosis, also called Kimmelstiel-Wilson lesions.

#138 · Renal Urinary and Reproductive Systems

How is nephritic syndrome defined?

Answer

Hematuria with dysmorphic red cells and red cell casts, some proteinuria, hypertension, and often a falling GFR, reflecting glomerular inflammation.

#139 · Renal Urinary and Reproductive Systems

What distinguishes IgA nephropathy from post-streptococcal glomerulonephritis?

Answer

IgA nephropathy causes gross hematuria within 1 to 2 days of an upper respiratory infection with normal complement. Post-strep GN has a 1 to 3 week latency and low complement.

#140 · Renal Urinary and Reproductive Systems

What immunofluorescence pattern is seen in anti-glomerular basement membrane (Goodpasture) disease?

Answer

A smooth, linear pattern of staining along the glomerular basement membrane, associated with pulmonary hemorrhage and rapidly progressive glomerulonephritis.

#141 • Renal Urinary and Reproductive Systems

Which class of lupus nephritis is the most aggressive, and what does biopsy show?

Answer

Class IV, diffuse proliferative lupus nephritis, showing more than half of glomeruli involved with subendothelial immune deposits; it requires intensive immunosuppression.

#142 • Renal Urinary and Reproductive Systems

What is the anion gap formula and normal range?

Answer

Anion gap equals sodium minus (chloride plus bicarbonate); normal is approximately 8 to 12 mEq/L.

#143 • Renal Urinary and Reproductive Systems

How fast can chronic hyponatremia safely be corrected, and why?

Answer

No faster than about 8 mEq/L in 24 hours, to avoid osmotic demyelination syndrome from too-rapid correction.

#144 • Renal Urinary and Reproductive Systems

What is the most common cause of euvolemic hyponatremia, and what cancer is classically linked to it?

Answer

Syndrome of inappropriate antidiuretic hormone secretion (SIADH); small cell lung cancer is a classic ectopic ADH source.

#145 • Renal Urinary and Reproductive Systems

What is the first treatment step for severe hyperkalemia with ECG changes?

Answer

Intravenous calcium gluconate, to stabilize the cardiac membrane, followed by insulin with dextrose to shift potassium intracellularly.

#146 • Renal Urinary and Reproductive Systems

What is the most common type of kidney stone, and what promotes it?

Answer

Calcium oxalate; promoted by low fluid intake and diets high in oxalate or animal protein.

#147 · Renal Urinary and Reproductive Systems

What organisms cause struvite stones, and why?

Answer

Urease-producing organisms such as *Proteus mirabilis*, which split urea into ammonia and alkalize the urine, allowing struvite crystals to form and grow into staghorn calculi.

#148 · Renal Urinary and Reproductive Systems

What stone type is radiolucent on plain X-ray but visible on CT?

Answer

Uric acid stones, which form in persistently acidic urine and are linked to gout and high-purine diets.

#149 • Renal Urinary and Reproductive Systems

When should asymptomatic bacteriuria be treated?

Answer

Only in pregnancy or before an invasive urologic procedure; otherwise treatment does not improve outcomes and promotes resistance.

#150 • Renal Urinary and Reproductive Systems

What are the four types of urinary incontinence and their basic mechanisms?

Answer

Stress (urethral hypermobility or sphincter weakness), urgency (overactive detrusor), overflow (underactive detrusor or obstruction with a large post-void residual), and functional (normal urinary tract, impaired access).

#151 • Renal Urinary and Reproductive Systems

What is first-line pharmacologic treatment for BPH, and why?

Answer

An alpha-1 blocker such as tamsulosin, because it relaxes prostatic and bladder neck smooth muscle for rapid symptom relief; finasteride shrinks the gland but takes months.

#152 • Renal Urinary and Reproductive Systems

What exam findings distinguish testicular torsion from epididymitis?

Answer

Torsion: high-riding testicle, horizontal lie, absent cremasteric reflex, sudden severe pain.
Epididymitis: gradual pain, relief with elevation (positive Prehn sign), intact cremasteric reflex.

#153 · Renal Urinary and Reproductive Systems

How do a hydrocele and a varicocele differ on exam?

Answer

A hydrocele transilluminates brightly and is generally painless. A varicocele feels like a bag of worms and decreases in size when the patient lies down.

#154 · Renal Urinary and Reproductive Systems

What are the Rotterdam criteria for polycystic ovary syndrome?

Answer

At least two of: oligo-ovulation or anovulation, clinical or biochemical hyperandrogenism, and polycystic ovaries on ultrasound, after excluding other causes of androgen excess.

#155 • Renal Urinary and Reproductive Systems

What is the classic symptom triad of endometriosis?

Answer

Cyclic pelvic pain beginning before menses, dyspareunia, and dyschezia; laparoscopy with visualization is the definitive diagnostic test.

#156 • Renal Urinary and Reproductive Systems

How does adenomyosis differ from uterine leiomyoma on exam?

Answer

Adenomyosis causes diffuse, tender uterine enlargement without a discrete mass. Leiomyoma (fibroid) causes an irregularly enlarged uterus with a discrete, well-circumscribed mass.

#157 · Renal Urinary and Reproductive Systems

What are the PALM-COEIN categories of abnormal uterine bleeding?

Answer

Structural: Polyp, Adenomyosis, Leiomyoma, Malignancy. Nonstructural: Coagulopathy, Ovulatory dysfunction, Endometrial, Iatrogenic, Not otherwise classified.

#158 · Renal Urinary and Reproductive Systems

How should a small, simple, thin-walled ovarian cyst in a premenopausal woman be managed?

Answer

Conservatively, with a repeat ultrasound in several weeks to months, since most are functional cysts that resolve on their own.

#159 · Renal Urinary and Reproductive Systems

What proportion of infertility involves a male factor, and what test evaluates it?

Answer

Roughly half of cases; semen analysis should be obtained early, in parallel with evaluation of the woman's ovulatory and anatomic factors.

#160 · Renal Urinary and Reproductive Systems

How do a fibroadenoma and fibrocystic breast changes typically present?

Answer

Fibroadenoma: a firm, mobile, well-circumscribed, painless mass, usually in young women. Fibrocystic changes: bilateral, cyclic tenderness and diffuse nodularity that fluctuates with the menstrual cycle.

#161 • Renal Urinary and Reproductive Systems

What breast finding requires prompt imaging and biopsy regardless of tenderness?

Answer

A new, firm, irregular, immobile mass, especially with skin dimpling, nipple retraction, or axillary lymphadenopathy, particularly in a woman over 40.

#162 • Renal Urinary and Reproductive Systems

What distinguishes inflammatory breast cancer from simple mastitis?

Answer

Inflammatory breast cancer causes erythema, warmth, and peau d'orange skin changes that do NOT improve with a full course of antibiotics, and requires skin punch biopsy.

#163 • Renal Urinary and Reproductive Systems

What is Paget disease of the breast, and what is it associated with?

Answer

An eczematous, crusting change of the nipple and areola that fails topical treatment, frequently associated with an underlying ductal carcinoma even without a palpable mass.

#164 • Renal Urinary and Reproductive Systems

Why is erectile dysfunction sometimes considered an early cardiovascular warning sign?

Answer

Small penile arteries develop atherosclerotic narrowing before larger coronary arteries do, so new ED can precede overt coronary artery disease.

#165 · Renal Urinary and Reproductive Systems

Why are PDE-5 inhibitors contraindicated with nitrate use?

Answer

Both increase nitric oxide-mediated vasodilation, and combining them can cause severe, potentially life-threatening hypotension.

#166 · Cardiovascular System

What ECG finding defines a STEMI?

Answer

ST elevation of at least 1 mm in two or more contiguous leads, or a new left bundle branch block with a compatible clinical history.

#167 · Cardiovascular System

Which coronary artery is usually occluded in an inferior STEMI, and what complication should be anticipated?

Answer

The right coronary artery; anticipate possible right ventricular infarction with preload-dependent hypotension, so avoid nitrates and morphine until this is clarified.

#168 · Cardiovascular System

What is the PCI time window that determines whether fibrinolysis should be given instead?

Answer

If primary PCI cannot be performed within 120 minutes of first medical contact (when transfer is needed), fibrinolytic therapy should be given within 30 minutes of arrival if not contraindicated.

#169 · Cardiovascular System

How are unstable angina and NSTEMI distinguished?

Answer

Both present with rest pain and possible ST depression or T wave inversion, but unstable angina has a normal troponin while NSTEMI has an elevated troponin.

#170 · Cardiovascular System

What mechanical complication of MI presents with a new holosystolic murmur at the apex and sudden flash pulmonary edema?

Answer

Papillary muscle rupture causing acute severe mitral regurgitation.

#171 · Cardiovascular System

When does free wall rupture with cardiac tamponade typically occur after MI, and how does it present?

Answer

Typically 3 to 5 days after a transmural infarct, presenting with sudden hypotension, jugular venous distension, and muffled heart sounds.

#172 · Cardiovascular System

What is Dressler syndrome and when does it occur?

Answer

An immune-mediated pericarditis occurring weeks after myocardial infarction, presenting with pleuritic pain relieved by sitting forward, low-grade fever, and diffuse ST elevation with PR depression.

#173 · Cardiovascular System

What defines HFrEF versus HFpEF?

Answer

HFrEF is heart failure with an ejection fraction of 40 percent or less; HFpEF is heart failure with an ejection fraction of 50 percent or greater.

#174 · Cardiovascular System

What are the four pillars of guideline directed medical therapy for HFrEF?

Answer

An ACE inhibitor or ARB (or ARNI), a beta blocker, a mineralocorticoid receptor antagonist, and an SGLT2 inhibitor.

#175 · Cardiovascular System

Which three beta blockers have proven mortality benefit in HFrEF?

Answer

Carvedilol, metoprolol succinate, and bisoprolol.

#176 · Cardiovascular System

What combination therapy has particular mortality benefit in Black patients with HFrEF who remain symptomatic on standard therapy?

Answer

Hydralazine plus isosorbide dinitrate.

#177 · Cardiovascular System

What does the bedside 'warm/cold, wet/dry' classification of heart failure guide?

Answer

Warm and wet patients generally need diuresis alone; cold and wet patients (poorly perfused) may need an inotrope in addition to diuresis.

#178 · Cardiovascular System

What does an S3 gallop reflect, and in what condition is it classically heard?

Answer

Rapid filling of a dilated, volume-overloaded ventricle during early diastole, classically heard in decompensated systolic heart failure (an S4, by contrast, reflects atrial contraction into a stiff, poorly compliant ventricle).

#179 · Cardiovascular System

What ECG finding characterizes atrial fibrillation?

Answer

An irregularly irregular rhythm with no discrete P waves.

#180 · Cardiovascular System

What score is used to guide anticoagulation decisions in atrial fibrillation, and what does a higher score mean?

Answer

The CHA₂DS₂-VASc score; a higher score indicates greater stroke risk and a stronger indication for anticoagulation.

#181 · Cardiovascular System

What is required before cardioversion of atrial fibrillation of unknown duration or over 48 hours?

Answer

Either at least 3 weeks of anticoagulation, or a transesophageal echocardiogram to exclude a left atrial thrombus.

#182 · Cardiovascular System

What is the first-line treatment for stable SVT that fails vagal maneuvers?

Answer

Adenosine, which transiently blocks AV node conduction and can terminate the reentrant circuit.

#183 · Cardiovascular System

How should an unstable tachyarrhythmia be treated regardless of the specific rhythm?

Answer

Immediate synchronized cardioversion.

#184 · Cardiovascular System

How should a symptomatic, unstable bradycardia be treated first?

Answer

Atropine; if that fails, transcutaneous pacing or an epinephrine or dopamine infusion while preparing for definitive pacing.

#185 · Cardiovascular System

What distinguishes Mobitz type I (Wenckebach) from Mobitz type II second degree AV block?

Answer

Mobitz I shows progressive PR prolongation before a dropped beat and is usually benign; Mobitz II shows a constant PR interval with unpredictably dropped beats and often requires a pacemaker due to risk of complete heart block.

#186 · Cardiovascular System

How is pulseless ventricular fibrillation treated?

Answer

Immediate high-quality chest compressions and unsynchronized defibrillation as soon as the shockable rhythm is confirmed.

#187 · Cardiovascular System

What murmur and symptom triad characterize severe aortic stenosis?

Answer

A harsh crescendo-decrescendo systolic murmur radiating to the carotids, with exertional syncope, angina, and heart failure symptoms as key warning signs.

#188 · Cardiovascular System

When is aortic valve replacement indicated in aortic stenosis?

Answer

Once symptoms develop in severe aortic stenosis, or when ejection fraction begins to fall, regardless of symptoms.

#189 · Cardiovascular System

Why does acute severe mitral regurgitation cause sudden pulmonary edema while chronic MR does not?

Answer

The left atrium has not had time to dilate and accommodate the new regurgitant volume in the acute setting, unlike in chronic MR where gradual dilation compensates for years.

#190 · Cardiovascular System

What is the classic auscultatory finding of mitral stenosis and its usual cause?

Answer

A low-pitched diastolic rumble at the apex with an opening snap, usually from rheumatic heart disease.

#191 · Cardiovascular System

Which anticoagulant is required in atrial fibrillation with moderate to severe mitral stenosis or a mechanical valve?

Answer

Warfarin, since direct oral anticoagulants are not proven safe or effective in these specific populations.

#192 · Cardiovascular System

What diagnostic framework combines blood cultures and echocardiographic findings for infective endocarditis?

Answer

The modified Duke criteria.

#193 · Cardiovascular System

At what blood pressure threshold is hypertension diagnosed?

Answer

Systolic 130 mmHg or higher, or diastolic 80 mmHg or higher, confirmed on at least two separate occasions.

#194 · Cardiovascular System

Which antihypertensive class is preferred first-line in a patient with diabetes and proteinuria or chronic kidney disease?

Answer

An ACE inhibitor or ARB, due to added benefit reducing intraglomerular pressure.

#195 · Cardiovascular System

What secondary hypertension clue suggests primary hyperaldosteronism?

Answer

Hypokalemia with metabolic alkalosis.

#196 · Cardiovascular System

What symptom triad suggests pheochromocytoma?

Answer

Episodic headache, palpitations, and diaphoresis with paroxysmal severe hypertension.

#197 · Cardiovascular System

What is the first medication given in a hypertensive emergency due to aortic dissection, and why?

Answer

An intravenous beta blocker, to reduce aortic wall shear stress (dP/dt) before adding a vasodilator, avoiding reflex tachycardia.

#198 · Cardiovascular System

Who should be screened for abdominal aortic aneurysm with a one-time ultrasound?

Answer

Men aged 65 to 75 who have ever smoked.

#199 · Cardiovascular System

What clinical clue suggests aortic dissection over other causes of chest pain?

Answer

Sudden, severe, tearing chest or back pain with a pulse deficit or a significant blood pressure difference between the arms.

#200 · Cardiovascular System

How does management differ between Stanford type A and type B aortic dissection?

Answer

Type A (involves the ascending aorta) is a surgical emergency; type B (confined to the descending aorta) is usually managed medically first unless there is malperfusion or another complication.

#201 · Cardiovascular System

How is the ankle-brachial index calculated, and what value is diagnostic of PAD?

Answer

The higher ankle systolic pressure divided by the higher brachial systolic pressure in each leg; a ratio below 0.9 is diagnostic of PAD.

#202 · Cardiovascular System

What are the 'six Ps' of acute limb ischemia?

Answer

Pain, pallor, poikilothermia, pulselessness, paresthesias, and paralysis.

#203 · Cardiovascular System

What distinguishes critical limb ischemia from simple claudication?

Answer

Critical limb ischemia includes rest pain, ulceration, or gangrene and requires urgent evaluation for revascularization.

#204 · Cardiovascular System

What features of syncope should prompt concern for a cardiac cause?

Answer

Syncope during exertion, syncope while supine, syncope without prodrome, syncope with palpitations or chest pain, a family history of sudden death, or an abnormal ECG.

#205 · Cardiovascular System

What ECG pattern is classic for acute pericarditis?

Answer

Diffuse ST elevation with PR segment depression, distinct from the localized ST elevation of myocardial infarction.

#206 · Musculoskeletal System and Skin

Colles fracture

Answer

A distal radius fracture with dorsal displacement of the distal fragment, from a fall on an outstretched hand, producing a silver-fork deformity.

#207 · Musculoskeletal System and Skin

Smith fracture

Answer

A distal radius fracture with volar displacement of the distal fragment, from a fall onto a flexed wrist, the mirror image of a Colles fracture.

#208 · Musculoskeletal System and Skin

Scaphoid fracture and avascular necrosis

Answer

The scaphoid's blood supply enters distally and travels retrograde, so proximal pole fractures risk avascular necrosis; snuffbox tenderness with a normal x-ray still warrants splinting and repeat imaging in 10 to 14 days.

#209 · Musculoskeletal System and Skin

Femoral neck fracture

Answer

A displaced fracture through the femoral neck risks avascular necrosis of the femoral head because the retrograde blood supply runs along the neck; urgent surgical fixation is required.

#210 · Musculoskeletal System and Skin

Supracondylar humerus fracture

Answer

A pediatric fracture from a fall onto an extended elbow that risks brachial artery and median nerve injury; a cool, pulseless hand requires emergent reduction.

#211 · Musculoskeletal System and Skin

Ottawa Ankle Rules

Answer

Obtain ankle radiographs if there is bone tenderness at the posterior edge or tip of either malleolus, or if the patient cannot bear weight for four steps immediately after injury and in the emergency department.

#212 · Musculoskeletal System and Skin

Compartment syndrome

Answer

Pain out of proportion to the injury and pain with passive stretch of the affected muscles are the earliest signs; emergent fasciotomy is the definitive treatment.

#213 · Musculoskeletal System and Skin

Volkmann contracture

Answer

Permanent flexion contracture of the forearm and hand resulting from unrecognized or inadequately treated compartment syndrome of the forearm.

#214 · Musculoskeletal System and Skin

Osteoarthritis

Answer

A mechanical joint disease from cartilage wear, causing brief morning stiffness under 30 minutes, pain that worsens with activity and improves with rest, and Heberden and Bouchard nodes.

#215 · Musculoskeletal System and Skin

Rheumatoid arthritis pattern

Answer

Symmetric small joint involvement of the wrists, MCP, and PIP joints, sparing the DIP joints, with morning stiffness over an hour that improves with activity.

#216 · Musculoskeletal System and Skin

Rheumatoid arthritis serology

Answer

Anti-cyclic citrullinated peptide antibody is more specific for rheumatoid arthritis than rheumatoid factor, though neither test alone confirms the diagnosis.

#217 · Musculoskeletal System and Skin

Rheumatoid arthritis treatment

Answer

Early initiation of a disease-modifying antirheumatic drug, most often methotrexate, prevents irreversible joint erosion; screen for latent tuberculosis and hepatitis B before starting a biologic agent.

#218 · Musculoskeletal System and Skin

Ankylosing spondylitis

Answer

Inflammatory back pain in a young patient that improves with exercise and worsens with rest, with sacroiliitis on imaging and a strong HLA-B27 association.

#219 · Musculoskeletal System and Skin

Dactylitis

Answer

Diffuse swelling of an entire digit, sometimes called a sausage digit, is a distinguishing feature of psoriatic arthritis.

#220 · Musculoskeletal System and Skin

Gout, acute presentation

Answer

Sudden, severe monoarticular pain, erythema, and swelling, classically at the first metatarsophalangeal joint, termed podagra.

#221 · Musculoskeletal System and Skin

Gout, crystal analysis

Answer

Needle-shaped, negatively birefringent monosodium urate crystals on polarized light microscopy confirm the diagnosis; synovial fluid should also be cultured to exclude septic arthritis.

#222 · Musculoskeletal System and Skin

Gout, urate-lowering therapy timing

Answer

Never start or adjust urate-lowering therapy during an acute flare; begin allopurinol only after the attack resolves, typically covered with colchicine or an NSAID for several months.

#223 · Musculoskeletal System and Skin

Gout, uric acid level pitfall

Answer

Serum uric acid can be normal or even fall during an acute flare, so it is an unreliable marker for diagnosing an attack in progress.

#224 · Musculoskeletal System and Skin

Pseudogout

Answer

Calcium pyrophosphate deposition disease affecting the knee or wrist in older adults, associated with hemochromatosis, hyperparathyroidism, hypothyroidism, and hypomagnesemia.

#225 · Musculoskeletal System and Skin

Pseudogout crystal analysis

Answer

Rhomboid-shaped, positively birefringent calcium pyrophosphate crystals, the opposite pattern from gout; radiographs may show chondrocalcinosis.

#226 · Musculoskeletal System and Skin

Septic arthritis rule-out

Answer

A hot, swollen, febrile joint always requires arthrocentesis with Gram stain and culture, since septic arthritis can mimic gout or an inflammatory flare and can coexist with crystal disease.

#227 · Musculoskeletal System and Skin

Osteoporosis screening

Answer

Dual-energy x-ray absorptiometry is recommended for all women starting at age 65, and earlier for younger postmenopausal women or men with significant risk factors.

#228 · Musculoskeletal System and Skin

T-score interpretation

Answer

A T-score of negative 2.5 or lower at the hip or spine defines osteoporosis; a T-score between negative 1.0 and negative 2.5 defines osteopenia.

#229 · Musculoskeletal System and Skin

Bisphosphonate use

Answer

Take on an empty stomach with a full glass of water and remain upright for 30 minutes; long-term use carries a small risk of atypical femoral fractures and osteonecrosis of the jaw.

#230 · Musculoskeletal System and Skin

Denosumab discontinuation risk

Answer

Abruptly stopping denosumab can cause a rebound increase in fracture risk, so transition to another antiresorptive agent is typically recommended.

#231 · Musculoskeletal System and Skin

Low back pain red flags

Answer

Age over 50 with new onset pain, cancer history, unexplained weight loss, fever, intravenous drug use, prolonged corticosteroid use, trauma, and progressive neurologic deficit all warrant further evaluation.

#232 · Musculoskeletal System and Skin

Cauda equina syndrome

Answer

Saddle anesthesia, bilateral leg weakness, and urinary retention with overflow incontinence; requires emergent MRI and surgical decompression.

#233 · Musculoskeletal System and Skin

Sciatica and straight leg raise

Answer

A positive straight leg raise test reproduces radicular pain below the knee at 30 to 70 degrees of elevation, supporting a herniated disc at L4-L5 or L5-S1.

#234 · Musculoskeletal System and Skin

L5 versus S1 radiculopathy

Answer

L5 radiculopathy causes weakness of great toe and foot dorsiflexion; S1 radiculopathy causes a diminished or absent ankle reflex.

#235 · Musculoskeletal System and Skin

Cellulitis versus erysipelas

Answer

Cellulitis produces poorly demarcated erythema and warmth; erysipelas, classically from *Streptococcus pyogenes*, has sharply demarcated, raised borders.

#236 · Musculoskeletal System and Skin

Necrotizing fasciitis warning signs

Answer

Pain out of proportion to visible findings, rapid progression, bullae, crepitus, and systemic toxicity should prompt emergent surgical consultation, since debridement is both diagnostic and therapeutic.

#237 · Musculoskeletal System and Skin

Psoriasis

Answer

Well-demarcated, erythematous plaques with thick silvery scale over extensor surfaces; the Auspitz sign is pinpoint bleeding when a scale is removed.

#238 · Musculoskeletal System and Skin

Psoriasis nail changes

Answer

Pitting, onycholysis, and oil-drop discoloration are common and can precede the development of psoriatic arthritis.

#239 · Musculoskeletal System and Skin

Atopic dermatitis

Answer

A chronic, pruritic condition associated with the atopic triad of asthma, allergic rhinitis, and eczema; favors flexural creases in older children and adults.

#240 · Musculoskeletal System and Skin

Basal cell carcinoma

Answer

The most common skin cancer, presenting as a pearly, translucent papule with rolled, telangiectatic borders; grows slowly and rarely metastasizes.

#241 · Musculoskeletal System and Skin

Squamous cell carcinoma

Answer

Arises from actinic keratosis, presenting as a firm, scaly, hyperkeratotic papule that may ulcerate; carries a modest metastatic risk, higher on the lip or ear.

#242 · Musculoskeletal System and Skin

Melanoma ABCDE criteria

Answer

Asymmetry, Border irregularity, Color variation, Diameter greater than 6 millimeters, and Evolution over time are warning signs requiring biopsy.

#243 · Musculoskeletal System and Skin

Melanoma biopsy technique

Answer

Full-thickness excisional biopsy, not shave biopsy, is required to accurately measure Breslow depth, the single most important prognostic factor.

#244 · Musculoskeletal System and Skin

Burn classification

Answer

Superficial burns involve only the epidermis. Superficial partial-thickness burns blister and are painful. Deep partial-thickness burns are pale or mottled with decreased sensation. Full-thickness burns are painless, leathery, and white or charred since nerve endings are destroyed.

#245 · Musculoskeletal System and Skin

Parkland formula

Answer

Total 24-hour fluid volume equals 4 mL times body weight in kilograms times percent total body surface area burned, with half given in the first 8 hours.

#246 · Behavioral Health and Psychiatry

What are the 8 SIG E CAPS symptoms of major depressive disorder, and how many total criteria are required for diagnosis?

Answer

Sleep change, Interest loss, Guilt/worthlessness, Energy loss, Concentration difficulty, Appetite/weight change, Psychomotor change, and Suicidal ideation. Depressed mood is a separate ninth criterion. At least 5 of these 9 total criteria are required for at least 2 weeks, and at least one must be depressed mood or anhedonia.

#247 · Behavioral Health and Psychiatry

What is the classic triad of serotonin syndrome and its most common precipitant?

Answer

Altered mental status, autonomic instability (fever, tachycardia, diaphoresis), and neuromuscular findings (hyperreflexia, clonus, worse in the lower extremities). It is most often precipitated by combining two serotonergic drugs, such as an SSRI with an MAOI, tramadol, or linezolid.

#248 · Behavioral Health and Psychiatry

How is bipolar I disorder distinguished from bipolar II disorder?

Answer

Bipolar I requires only one full manic episode (at least 1 week, marked impairment or hospitalization). Bipolar II requires a hypomanic episode (at least 4 days, no marked impairment) plus at least one major depressive episode.

#249 · Behavioral Health and Psychiatry

Why should SSRIs generally be avoided as monotherapy in bipolar depression?

Answer

An antidepressant given without a mood stabilizer or antipsychotic can precipitate a manic or hypomanic switch. Mood stabilizers (lithium, valproate, lamotrigine) or atypical antipsychotics are preferred first.

#250 · Behavioral Health and Psychiatry

What labs should be monitored regularly in a patient on lithium, and what worsens lithium clearance?

Answer

Renal function, thyroid function, and serum lithium level should be monitored regularly. Dehydration, NSAIDs, ACE inhibitors, and thiazide diuretics all reduce renal lithium clearance and raise toxicity risk.

#251 · Behavioral Health and Psychiatry

What is the major teratogenic risk of valproate, and what lab requires monitoring?

Answer

Valproate causes neural tube defects and is generally avoided in women of childbearing potential when possible. Liver function tests should be monitored because of hepatotoxicity risk.

#252 · Behavioral Health and Psychiatry

Why must lamotrigine be titrated slowly?

Answer

Rapid dose escalation increases the risk of Stevens-Johnson syndrome, a severe and potentially life-threatening skin reaction.

#253 · Behavioral Health and Psychiatry

What are the diagnostic criteria for generalized anxiety disorder?

Answer

Excessive, hard-to-control worry about multiple life domains for at least 6 months, with at least 3 of 6 associated symptoms: restlessness, fatigue, poor concentration, irritability, muscle tension, and sleep disturbance.

#254 · Behavioral Health and Psychiatry

What defines panic disorder, as opposed to an isolated panic attack?

Answer

Recurrent unexpected panic attacks (abrupt fear peaking within minutes with palpitations, sweating, and doom) plus at least 1 month of persistent worry about future attacks or avoidance behavior to prevent them.

#255 · Behavioral Health and Psychiatry

What is first-line treatment across the anxiety disorders, and what role do benzodiazepines play?

Answer

Cognitive behavioral therapy combined with an SSRI or SNRI is first-line. Benzodiazepines can relieve acute anxiety quickly but are not first-line for chronic management because of dependence, sedation, and fall risk.

#256 · Behavioral Health and Psychiatry

What is buspirone's key advantage and key limitation as an anxiolytic?

Answer

It is non-sedating and lacks abuse potential, but it takes weeks to become effective and has no role in acute panic relief.

#257 · Behavioral Health and Psychiatry

How long must symptoms persist to distinguish acute stress disorder from PTSD?

Answer

Acute stress disorder lasts between 3 days and 1 month after the trauma. If the same symptom pattern persists beyond 1 month, the diagnosis becomes PTSD.

#258 · Behavioral Health and Psychiatry

What is the minimum symptom duration to diagnose schizophrenia, and how does this differ from schizophreniform disorder and brief psychotic disorder?

Answer

Schizophrenia requires at least 6 months of continuous disturbance. Schizophreniform disorder is the same symptom picture lasting 1 to 6 months. Brief psychotic disorder lasts less than 1 month.

#259 · Behavioral Health and Psychiatry

How is schizoaffective disorder distinguished from a mood disorder with psychotic features?

Answer

Schizoaffective disorder requires at least 2 weeks of delusions or hallucinations in the absence of a mood episode at some point. In mood disorder with psychotic features, psychosis occurs only during mood episodes.

#260 · Behavioral Health and Psychiatry

List the timeline of antipsychotic-induced movement side effects from earliest to latest.

Answer

Acute dystonia (hours to days), akathisia (days to weeks), parkinsonism (weeks), and tardive dyskinesia (months to years), which may be irreversible.

#261 · Behavioral Health and Psychiatry

What is the treatment for acute dystonia versus akathisia?

Answer

Acute dystonia is treated with an anticholinergic such as benztropine or diphenhydramine. Akathisia is treated by lowering the antipsychotic dose or adding a beta-blocker or benzodiazepine.

#262 · Behavioral Health and Psychiatry

What is the tetrad of neuroleptic malignant syndrome and its treatment?

Answer

Fever, lead-pipe rigidity, autonomic instability, and altered mental status, with markedly elevated creatine kinase. Treatment is immediate antipsychotic discontinuation, supportive cooling, and dantrolene or bromocriptine for severe cases.

#263 · Behavioral Health and Psychiatry

Why is clozapine reserved for treatment-resistant schizophrenia, and what monitoring does it require?

Answer

It carries a risk of agranulocytosis, so it is used only after failure of at least two adequate antipsychotic trials and requires mandatory regular absolute neutrophil count monitoring.

#264 · Behavioral Health and Psychiatry

What are the defining features of borderline personality disorder?

Answer

Instability of relationships, self-image, and affect; marked impulsivity; chronic emptiness; recurrent suicidal or self-harm behavior; and an intense fear of abandonment. First-line treatment is dialectical behavior therapy.

#265 · Behavioral Health and Psychiatry

How does schizoid personality disorder differ from avoidant personality disorder?

Answer

Schizoid personality disorder involves preference for solitude with no distress about isolation. Avoidant personality disorder involves social inhibition driven by fear of criticism and feelings of inadequacy, with the person wanting connection but avoiding it out of fear.

#266 · Behavioral Health and Psychiatry

What are the three DSM-5-style personality disorder clusters, with one example disorder each?

Answer

Cluster A, odd/eccentric (e.g., paranoid); Cluster B, dramatic/emotional/erratic (e.g., borderline); Cluster C, anxious/fearful (e.g., avoidant).

#267 · Behavioral Health and Psychiatry

What is the timeline for alcohol withdrawal seizures versus delirium tremens?

Answer

Withdrawal seizures can occur 12 to 48 hours after the last drink. Delirium tremens, the most severe form, occurs 48 to 96 hours after the last drink.

#268 · Behavioral Health and Psychiatry

Why is thiamine given before glucose in a patient with suspected alcohol withdrawal?

Answer

Giving glucose first in a thiamine-deficient patient can precipitate or worsen Wernicke encephalopathy by consuming remaining thiamine stores during glucose metabolism.

#269 · Behavioral Health and Psychiatry

What is the classic triad of Wernicke encephalopathy?

Answer

Confusion, ataxia, and ophthalmoplegia (gaze palsy), classically in a chronically alcohol-dependent, malnourished patient.

#270 · Behavioral Health and Psychiatry

What happens when naloxone is given to an opioid-dependent patient, and what are the symptoms produced?

Answer

It can precipitate acute opioid withdrawal: dilated pupils, yawning, rhinorrhea, lacrimation, piloerection, myalgias, abdominal cramping, and diarrhea. It should be titrated carefully in dependent patients.

#271 · Behavioral Health and Psychiatry

Why should benzodiazepines never be stopped abruptly after long-term use?

Answer

Abrupt discontinuation can precipitate a withdrawal syndrome resembling alcohol withdrawal, including seizure risk. A gradual taper, sometimes with substitution of a longer-acting agent, is required.

#272 · Behavioral Health and Psychiatry

What is the key difference between malingering and factitious disorder?

Answer

Malingering involves conscious symptom fabrication for a clear external incentive (financial gain, avoiding work, obtaining opioids). Factitious disorder involves conscious falsification with no external incentive, motivated by a wish to assume the sick role itself.

#273 · Behavioral Health and Psychiatry

How does conversion disorder differ from somatic symptom disorder?

Answer

Conversion disorder involves specific neurological symptoms incompatible with recognized neurological disease, often following a stressor. Somatic symptom disorder involves distressing somatic symptoms of any kind paired with excessive worry, thoughts, or behavior about them for over 6 months.

#274 · Behavioral Health and Psychiatry

What electrolyte and cardiac complication defines refeeding syndrome, and how is it prevented?

Answer

An insulin surge from reintroduced nutrition drives phosphate, potassium, and magnesium into cells, causing hypophosphatemia that can lead to cardiac and respiratory failure. It is prevented by advancing calories slowly with proactive electrolyte monitoring and repletion.

#275 · Behavioral Health and Psychiatry

How does bulimia nervosa differ from anorexia nervosa, binge-eating/purging type?

Answer

Bulimia nervosa occurs at a normal or above-normal body weight. Anorexia nervosa, even the binge-eating/purging subtype, requires a significantly low body weight.

#276 · Behavioral Health and Psychiatry

What is the single strongest risk factor for completed suicide?

Answer

A prior suicide attempt is the single strongest predictor of a future completed suicide.

#277 · Behavioral Health and Psychiatry

What is the Tarasoff principle, and when does it apply?

Answer

When a patient makes a credible, specific threat toward an identifiable person, the clinician has a duty to warn or protect that person and notify appropriate authorities, overriding usual confidentiality.

#278 · Behavioral Health and Psychiatry

How is catatonia treated, and what confirms the diagnosis?

Answer

A benzodiazepine trial, commonly lorazepam, is first-line and is also diagnostic when it produces rapid improvement. Electroconvulsive therapy is used for benzodiazepine-refractory catatonia.

#279 · Behavioral Health and Psychiatry

What key features distinguish delirium from dementia?

Answer

Delirium has an acute onset, a fluctuating course, and impaired attention, usually from an underlying medical cause. Dementia has an insidious onset and a stable, progressive course, typically with attention preserved early on.

#280 · Behavioral Health and Psychiatry

What are the main indications for electroconvulsive therapy, and its main side effect?

Answer

Severe or treatment-resistant depression, depression with psychotic features, catatonia, and situations needing rapid response (e.g., an actively suicidal or severely malnourished patient). Its main side effect is transient anterograde and retrograde memory impairment.

#281 · Blood Lymphoreticular and Immune Systems

What is the classic iron study pattern (ferritin, iron, TIBC) in iron deficiency anemia versus anemia of chronic disease?

Answer

Iron deficiency: low ferritin, low iron, high TIBC.
Anemia of chronic disease: normal or high ferritin, low iron, low TIBC.

#282 · Blood Lymphoreticular and Immune Systems

What is the next best step when an older adult presents with new, unexplained iron deficiency anemia?

Answer

Colonoscopy (and often upper endoscopy) to evaluate for an occult gastrointestinal source of blood loss, including colorectal cancer.

#283 · Blood Lymphoreticular and Immune Systems

What laboratory and smear findings distinguish thalassemia trait from iron deficiency anemia?

Answer

Thalassemia trait has a normal iron panel, a normal RDW, and a normal or high red cell count for the degree of microcytosis, unlike iron deficiency.

#284 · Blood Lymphoreticular and Immune Systems

What neurologic finding is unique to vitamin B12 deficiency and never seen with folate deficiency?

Answer

Subacute combined degeneration of the spinal cord, involving both the dorsal columns and corticospinal tracts.

#285 · Blood Lymphoreticular and Immune Systems

What smear finding confirms megaloblastic anemia regardless of whether the cause is B12 or folate deficiency?

Answer

Hypersegmented neutrophils.

#286 · Blood Lymphoreticular and Immune Systems

What is the most likely cause of a normocytic anemia with a low reticulocyte count in advanced chronic kidney disease?

Answer

Reduced renal erythropoietin production.

#287 · Blood Lymphoreticular and Immune Systems

What single test best distinguishes immune from nonimmune causes of hemolytic anemia?

Answer

The direct antiglobulin (Coombs) test.

#288 · Blood Lymphoreticular and Immune Systems

What antibody class and clinical trigger characterize warm autoimmune hemolytic anemia versus cold agglutinin disease?

Answer

Warm AIHA is IgG-mediated and associated with lymphoproliferative disease or lupus. Cold agglutinin disease is IgM-mediated and triggered by cold exposure, often after *Mycoplasma pneumoniae* infection.

#289 · Blood Lymphoreticular and Immune Systems

What peripheral smear finding is classic for hereditary spherocytosis, and what is the definitive treatment for severe disease?

Answer

Spherocytes on smear (with a positive osmotic fragility test); splenectomy is the definitive treatment.

#290 · Blood Lymphoreticular and Immune Systems

What enzyme is deficient in G6PD deficiency, and what smear findings result from an oxidative trigger?

Answer

Glucose-6-phosphate dehydrogenase; bite cells and Heinz bodies appear after oxidative stress from infection, sulfa drugs, or fava beans.

#291 · Blood Lymphoreticular and Immune Systems

What four conditions classically cause microangiopathic hemolytic anemia with schistocytes on smear?

Answer

Thrombotic thrombocytopenic purpura, hemolytic uremic syndrome, disseminated intravascular coagulation, and mechanical heart valves.

#292 · Blood Lymphoreticular and Immune Systems

What surface proteins are lost in paroxysmal nocturnal hemoglobinuria, and what unusual thrombosis site is classic for this disease?

Answer

CD55 and CD59; hepatic vein (Budd-Chiari) thrombosis is a classic unusual-site clot.

#293 · Blood Lymphoreticular and Immune Systems

What smear and functional finding reflects functional asplenia in sickle cell disease?

Answer

Howell-Jolly bodies on smear, reflecting loss of splenic clearance function from repeated splenic infarction.

#294 · Blood Lymphoreticular and Immune Systems

What cytogenetic abnormality and cellular finding define acute promyelocytic leukemia, and why is it a hematologic emergency?

Answer

The t(15;17) translocation with Auer rods on smear; it can trigger disseminated intravascular coagulation at presentation and is treated urgently with all-trans retinoic acid.

#295 · Blood Lymphoreticular and Immune Systems

What is the Philadelphia chromosome, and what drug class targets the protein it produces?

Answer

The t(9;22) translocation producing the BCR-ABL fusion protein, targeted by tyrosine kinase inhibitors such as imatinib in chronic myeloid leukemia.

#296 · Blood Lymphoreticular and Immune Systems

What peripheral smear finding is classic for chronic lymphocytic leukemia?

Answer

Smudge cells, from fragile malignant lymphocytes damaged during slide preparation.

#297 · Blood Lymphoreticular and Immune Systems

What is the diagnostic cell of Hodgkin lymphoma, and what is its characteristic appearance?

Answer

The Reed-Sternberg cell, a large binucleate cell with an owl-eye appearance.

#298 · Blood Lymphoreticular and Immune Systems

What translocation and oncogene drive Burkitt lymphoma, and what is its classic histologic appearance?

Answer

The t(8;14) translocation activating c-myc; a starry sky pattern of tingible body macrophages among rapidly dividing lymphoid cells.

#299 · Blood Lymphoreticular and Immune Systems

What translocation and gene are implicated in follicular lymphoma, and what does this gene normally do?

Answer

The t(14;18) translocation causing BCL-2 overexpression; BCL-2 normally inhibits apoptosis, so its overexpression lets malignant cells evade cell death.

#300 · Blood Lymphoreticular and Immune Systems

What is the classic mnemonic for the clinical features of multiple myeloma, and what causes its renal impairment?

Answer

CRAB: hypercalcemia, renal failure, anemia, bone lesions; renal impairment comes from filtered free light chains causing cast nephropathy, compounded by hypercalcemia.

#301 · Blood Lymphoreticular and Immune Systems

What mutation drives most cases of polycythemia vera, and what is its main clinical risk?

Answer

A JAK2 mutation; the main risk is arterial and venous thrombosis from increased blood viscosity.

#302 · Blood Lymphoreticular and Immune Systems

What smear finding and marrow finding characterize primary myelofibrosis?

Answer

A leukoerythroblastic smear with teardrop cells; a dry tap on marrow aspiration due to extensive fibrosis.

#303 · Blood Lymphoreticular and Immune Systems

How do PT and aPTT localize a bleeding disorder to primary versus secondary hemostasis?

Answer

Mucocutaneous bleeding with a normal PT/aPTT suggests primary hemostasis (platelets, von Willebrand factor); deep bleeding into joints with a prolonged PT and/or aPTT suggests a factor deficiency in secondary hemostasis.

#304 · Blood Lymphoreticular and Immune Systems

Why can von Willebrand disease prolong the aPTT even though it is a primary hemostasis disorder?

Answer

Von Willebrand factor stabilizes circulating factor VIII, so its deficiency can secondarily lower factor VIII and prolong the aPTT.

#305 · Blood Lymphoreticular and Immune Systems

What is the inheritance pattern of hemophilia A and B, and what coagulation study is prolonged?

Answer

X-linked recessive; the aPTT is prolonged while the PT stays normal.

#306 · Blood Lymphoreticular and Immune Systems

What four laboratory findings define disseminated intravascular coagulation?

Answer

Prolonged PT and aPTT, low platelets, low fibrinogen, and elevated D-dimer.

#307 · Blood Lymphoreticular and Immune Systems

Why does antiphospholipid antibody syndrome prolong the aPTT in vitro despite causing thrombosis in vivo?

Answer

The antiphospholipid antibodies interfere with the phospholipid-dependent reagent used in the aPTT assay, a laboratory artifact unrelated to the true prothrombotic clinical picture.

#308 · Blood Lymphoreticular and Immune Systems

Why must warfarin be started with an overlapping heparin bridge in a patient with protein C or S deficiency?

Answer

Protein C and S are vitamin K-dependent anticoagulants with shorter half-lives than the procoagulant factors, so warfarin alone can transiently worsen hypercoagulability and cause skin necrosis.

#309 · Blood Lymphoreticular and Immune Systems

What is the immediate first step in managing a suspected acute hemolytic transfusion reaction?

Answer

Stop the transfusion immediately, then provide aggressive intravenous fluid resuscitation.

#310 · Blood Lymphoreticular and Immune Systems

What distinguishes transfusion-related acute lung injury from transfusion-associated circulatory overload?

Answer

TRALI causes hypotension with noncardiogenic pulmonary edema from donor antibody-mediated neutrophil activation; TACO causes hypertension and volume overload from excessive transfusion volume or rate.

#311 · Blood Lymphoreticular and Immune Systems

Which patients are at risk for anaphylactic transfusion reactions, and how are future reactions prevented?

Answer

Patients with selective IgA deficiency who have anti-IgA antibodies; future transfusions use washed blood products with plasma proteins removed.

#312 · Blood Lymphoreticular and Immune Systems

What CD4 thresholds trigger prophylaxis against *Pneumocystis jirovecii*, *Toxoplasma gondii*, and *Mycobacterium avium* complex?

Answer

Below 200 for *Pneumocystis jirovecii*, below 100 for *Toxoplasma gondii*, and below 50 for *Mycobacterium avium* complex.

#313 · Blood Lymphoreticular and Immune Systems

What immune defect explains recurrent *Neisseria* infections with normal immunoglobulin levels?

Answer

A deficiency of a terminal complement component, C5 through C9, which impairs membrane attack complex formation.

#314 · Blood Lymphoreticular and Immune Systems

What defect underlies chronic granulomatous disease, and how is it diagnosed?

Answer

A defective neutrophil NADPH oxidase impairing the oxidative burst, causing recurrent infection with catalase-positive organisms; diagnosed with an abnormal nitroblue tetrazolium or dihydrorhodamine test.

#315 · Blood Lymphoreticular and Immune Systems

What are the four types of hypersensitivity reactions and one classic example of each?

Answer

Type I: IgE-mediated, anaphylaxis. Type II: antibody-mediated cytotoxicity, Goodpasture syndrome. Type III: immune complex-mediated, serum sickness. Type IV: delayed T cell-mediated, contact dermatitis.

#316 · Gastrointestinal System

What alarm features mandate upper endoscopy before empiric PPI therapy in a patient with reflux symptoms?

Answer

Dysphagia, odynophagia, unintentional weight loss, GI bleeding or anemia, and age over 60 at symptom onset.

#317 · Gastrointestinal System

What is Barrett esophagus and what does it increase the risk of?

Answer

Intestinal metaplasia (columnar epithelium with goblet cells) of the distal esophagus from chronic reflux, which increases the risk of esophageal adenocarcinoma.

#318 · Gastrointestinal System

Why do duodenal ulcers not require routine follow-up endoscopy while gastric ulcers do?

Answer

Duodenal ulcers are almost never malignant, but gastric ulcers can look identical to malignancy at initial inspection, so repeat endoscopy in six to eight weeks is required to confirm healing and exclude cancer.

#319 · Gastrointestinal System

What findings suggest Zollinger-Ellison syndrome in a patient with peptic ulcer disease?

Answer

Multiple, refractory, or unusually distal ulcers with chronic diarrhea and a markedly elevated fasting serum gastrin level, from a gastrin-secreting tumor.

#320 · Gastrointestinal System

Why does an upper GI bleed raise BUN out of proportion to creatinine?

Answer

Blood protein sitting in the small bowel is digested and its nitrogen is absorbed, raising BUN while creatinine stays relatively unchanged.

#321 · Gastrointestinal System

What empiric therapy should be started before endoscopy in suspected variceal hemorrhage?

Answer

Intravenous octreotide to reduce splanchnic blood flow and an antibiotic such as ceftriaxone to reduce the risk of spontaneous bacterial peritonitis and rebleeding.

#322 · Gastrointestinal System

What is the preferred endoscopic technique for actively bleeding esophageal varices?

Answer

Band ligation, which has replaced sclerotherapy as first-line therapy due to a lower complication rate.

#323 · Gastrointestinal System

When is TIPS indicated in variceal bleeding?

Answer

When bleeding continues or recurs despite adequate endoscopic therapy, TIPS decompresses the portal system.

#324 · Gastrointestinal System

What is Heyde syndrome?

Answer

The association of aortic stenosis with bleeding colonic angiodysplasia, thought to result from acquired von Willebrand factor deficiency caused by shear stress across the stenotic valve.

#325 · Gastrointestinal System

What is the classic anatomic and histologic difference between Crohn disease and ulcerative colitis?

Answer

Crohn disease can affect any part of the GI tract with skip lesions and transmural inflammation with granulomas; ulcerative colitis is continuous from the rectum proximally, limited to the colon, with mucosal and submucosal inflammation and no granulomas.

#326 · Gastrointestinal System

Which extraintestinal manifestation of IBD is specifically linked to ulcerative colitis and tends to track independently of bowel disease activity?

Answer

Primary sclerosing cholangitis.

#327 · Gastrointestinal System

Why is 5-aminosalicylate therapy less useful in Crohn disease than in ulcerative colitis?

Answer

5-ASA agents have limited efficacy in Crohn disease, so Crohn disease often requires escalation to corticosteroids, immunomodulators, or biologics instead.

#328 · Gastrointestinal System

When is surgery curative in IBD, and when is it not?

Answer

Total colectomy is curative for ulcerative colitis since disease is confined to the colon. Surgery in Crohn disease treats complications like strictures or fistulas but is not curative since disease can recur anywhere in the GI tract.

#329 · Gastrointestinal System

When should colorectal cancer surveillance begin in long-standing IBD?

Answer

Approximately eight to ten years after diagnosis, given rising colorectal cancer risk with duration and extent of colonic involvement.

#330 · Gastrointestinal System

How do you interpret hepatitis B surface antigen positive with surface antibody negative?

Answer

This indicates active infection, whether acute or chronic, since surface antigen reflects ongoing viral presence.

#331 · Gastrointestinal System

How do you distinguish vaccination from resolved natural hepatitis B infection serologically?

Answer

Vaccination produces surface antibody positivity with a negative core antibody. Resolved natural infection produces both surface antibody and core antibody positivity.

#332 · Gastrointestinal System

Who should be treated for chronic hepatitis C infection?

Answer

Essentially all patients with chronic hepatitis C regardless of fibrosis stage, using direct acting antiviral therapy.

#333 · Gastrointestinal System

What AST to ALT pattern suggests alcoholic hepatitis?

Answer

An AST to ALT ratio greater than 2, distinct from most other causes of transaminase elevation where ALT typically exceeds AST.

#334 · Gastrointestinal System

What five variables make up the Child-Pugh classification?

Answer

Bilirubin, albumin, INR, ascites, and hepatic encephalopathy, used to stratify cirrhosis severity and estimate surgical risk.

#335 · Gastrointestinal System

What is the MELD score used for?

Answer

Prioritizing patients for liver transplantation, calculated from bilirubin, creatinine, and INR.

#336 · Gastrointestinal System

How is the serum-ascites albumin gradient interpreted?

Answer

A gradient of 1.1 g/dL or greater indicates a portal hypertensive cause such as cirrhosis; a gradient below 1.1 g/dL suggests another cause such as peritoneal carcinomatosis.

#337 · Gastrointestinal System

What ascitic fluid neutrophil count diagnoses spontaneous bacterial peritonitis, and what is added to antibiotics in high-risk patients?

Answer

An absolute neutrophil count of 250 cells/mm³ or higher. Albumin infusion is added in patients with significant renal or liver dysfunction to reduce hepatorenal syndrome risk.

#338 · Gastrointestinal System

What is the mechanism and first-line treatment of hepatic encephalopathy?

Answer

Accumulation of gut-derived ammonia the cirrhotic liver cannot clear; first-line treatment is lactulose, which acidifies the colon and traps ammonia as ammonium for excretion.

#339 · Gastrointestinal System

What are the diagnostic criteria for acute pancreatitis?

Answer

At least 2 of 3: characteristic epigastric pain radiating to the back, lipase or amylase at least 3 times the upper limit of normal, and characteristic findings on imaging. Not all three are required.

#340 · Gastrointestinal System

Why is early enteral feeding preferred over prolonged bowel rest in acute pancreatitis?

Answer

Early oral or enteral nutrition as tolerated improves outcomes compared to prolonged fasting and parenteral nutrition.

#341 · Gastrointestinal System

What distinguishes biliary colic from acute cholecystitis?

Answer

Biliary colic is transient pain from temporary cystic duct obstruction that resolves completely without inflammation. Acute cholecystitis is persistent obstruction with gallbladder wall inflammation, fever, and a positive Murphy sign.

#342 · Gastrointestinal System

What is the Charcot triad and the Reynolds pentad?

Answer

Charcot triad is fever, jaundice, and right upper quadrant pain, seen in ascending cholangitis. Reynolds pentad adds hypotension and altered mental status, indicating more severe cholangitis.

#343 · Gastrointestinal System

What is acalculous cholecystitis and who is at risk?

Answer

Gallbladder inflammation without gallstones, from bile stasis and ischemia, seen in critically ill patients with prolonged fasting or mechanical ventilation; carries a higher risk of gangrene and perforation.

#344 · Gastrointestinal System

What is the classic pain migration pattern of appendicitis?

Answer

Periumbilical pain that migrates to the right lower quadrant over hours, with tenderness at McBurney point, anorexia, nausea, and low-grade fever.

#345 · Gastrointestinal System

Why should colonoscopy be avoided during acute diverticulitis but performed six to eight weeks later?

Answer

During the acute episode it carries a perforation risk; afterward it is used to exclude an underlying colorectal malignancy.

#346 · Gastrointestinal System

What is the most common cause of small bowel obstruction versus large bowel obstruction in adults?

Answer

Small bowel obstruction is most often from adhesions after prior surgery. Large bowel obstruction is more often from colorectal malignancy, volvulus, or diverticular stricture.

#347 · Gastrointestinal System

At what age should average-risk colorectal cancer screening begin, and what modalities are accepted?

Answer

Age 45, using colonoscopy every ten years, annual stool-based testing such as FIT, or CT colonography at an appropriate interval.

#348 · Gastrointestinal System

When should colorectal cancer screening begin in a patient with a first-degree relative diagnosed before age 60?

Answer

At age 40 or ten years before the relative's age at diagnosis, whichever comes first.

#349 · Gastrointestinal System

What is the preferred initial serologic test for celiac disease and why must total IgA also be checked?

Answer

IgA tissue transglutaminase antibody; total IgA is checked because selective IgA deficiency causes a false negative tissue transglutaminase result.

#350 · Gastrointestinal System

Why should gluten not be stopped before duodenal biopsy for suspected celiac disease?

Answer

Early gluten avoidance can normalize both serology and villous histology, producing a false negative biopsy and obscuring the diagnosis.

#351 · Nervous System and Special Senses

What is the first imaging test in any patient with acute focal neurologic deficits?

Answer

Noncontrast head CT, to rapidly exclude hemorrhage before considering thrombolysis or thrombectomy.

#352 · Nervous System and Special Senses

Middle cerebral artery stroke: which body parts are most affected?

Answer

Contralateral face and arm weakness greater than leg weakness, plus aphasia (dominant hemisphere) or neglect (nondominant hemisphere).

#353 · Nervous System and Special Senses

Anterior cerebral artery stroke: which body part is most affected?

Answer

Contralateral leg weakness greater than arm weakness, since the leg motor cortex is on the medial surface supplied by this artery.

#354 · Nervous System and Special Senses

Posterior cerebral artery stroke: classic visual finding?

Answer

Contralateral homonymous hemianopia with macular sparing.

#355 • Nervous System and Special Senses

What is the IV thrombolysis time window for acute ischemic stroke?

Answer

Within 4.5 hours of clearly known symptom onset, if no contraindications are present.

#356 • Nervous System and Special Senses

Classic CT finding and cause of a basal ganglia intracerebral hemorrhage?

Answer

Hyperdense lesion on noncontrast CT, most often from chronic hypertension damaging small penetrating vessels.

#357 · Nervous System and Special Senses

Subarachnoid hemorrhage: classic history and confirmatory test if CT is negative?

Answer

Thunderclap headache reaching maximum intensity within seconds; if CT is negative, lumbar puncture looking for xanthochromia.

#358 · Nervous System and Special Senses

What is the most feared delayed complication of subarachnoid hemorrhage, and when does it occur?

Answer

Cerebral vasospasm, typically 3 to 14 days after the bleed, treated or prevented with nimodipine.

#359 · Nervous System and Special Senses

Glasgow Coma Scale components and the threshold for airway concern?

Answer

Eye opening (4), verbal response (5), motor response (6); a total score of 8 or less generally prompts consideration of intubation.

#360 · Nervous System and Special Senses

Definition of a transient ischemic attack (TIA)?

Answer

Focal neurologic deficits that resolve completely, typically within an hour, with no infarct on imaging.

#361 • Nervous System and Special Senses

Diagnostic criteria for epilepsy after a first seizure?

Answer

Two unprovoked seizures more than 24 hours apart, or one unprovoked seizure with a high recurrence risk on imaging or EEG.

#362 • Nervous System and Special Senses

Which antiepileptic drugs are broad spectrum, effective for both focal and generalized seizures?

Answer

Levetiracetam and valproic acid.

#363 · Nervous System and Special Senses

Why is valproic acid generally avoided in women of childbearing potential?

Answer

High risk of neural tube defects and other major fetal malformations.

#364 · Nervous System and Special Senses

Definition of status epilepticus and first line treatment?

Answer

A single seizure lasting 5 minutes or more, or recurrent seizures without recovery between them; first line treatment is a benzodiazepine.

#365 • Nervous System and Special Senses

Classic cerebrospinal fluid finding of albuminocytologic dissociation is seen in which disease?

Answer

Guillain-Barre syndrome (elevated protein with a normal white blood cell count).

#366 • Nervous System and Special Senses

What distinguishes psychogenic nonepileptic seizures from true epileptic seizures?

Answer

Asynchronous thrashing, pelvic thrusting, forced eye closure, waxing and waning course, and a normal EEG during the event.

#367 · Nervous System and Special Senses

Migraine diagnostic features?

Answer

Unilateral, pulsatile headache lasting 4 to 72 hours with nausea, photophobia, phonophobia, and worsening with activity.

#368 · Nervous System and Special Senses

Cluster headache: classic presentation and acute treatment?

Answer

Severe unilateral periorbital pain lasting 15 minutes to 3 hours with ipsilateral autonomic features; treated acutely with high flow oxygen or a triptan.

#369 · Nervous System and Special Senses

What does the SNOOP mnemonic screen for in headache patients?

Answer

Red flags for secondary headache: Systemic symptoms, Neurologic signs, Onset sudden (thunderclap), Older age of onset, Pattern change/progressive.

#370 · Nervous System and Special Senses

Headache red flag suggesting giant cell arteritis, and the urgent treatment?

Answer

New headache after age 50 with jaw claudication and visual symptoms; treat with empiric high dose glucocorticoids immediately to prevent vision loss.

#371 • Nervous System and Special Senses

Idiopathic intracranial hypertension: typical patient and treatment?

Answer

Obese woman of childbearing age with headache, papilledema, and normal imaging aside from possibly an empty sella; treated with acetazolamide and weight loss.

#372 • Nervous System and Special Senses

Delirium vs dementia: onset and course?

Answer

Delirium: acute onset (hours to days), fluctuating course, core deficit in attention. Dementia: insidious onset (months to years), stable course, core deficit in memory.

#373 · Nervous System and Special Senses

Alzheimer disease: earliest and most prominent feature, and classic imaging finding?

Answer

Short-term memory loss first, followed by language and visuospatial impairment; imaging shows cortical atrophy with hippocampal predominance.

#374 · Nervous System and Special Senses

Lewy body dementia: core triad and a key management caution?

Answer

Fluctuating cognition, visual hallucinations, and parkinsonism; avoid antipsychotics because of severe sensitivity reactions.

#375 · Nervous System and Special Senses

Normal pressure hydrocephalus: classic triad and a test that predicts shunt response?

Answer

Gait disturbance, urinary incontinence, and cognitive decline; gait improvement after large volume lumbar puncture predicts response to shunting.

#376 · Nervous System and Special Senses

Multiple sclerosis: core diagnostic concept and a classic exam finding?

Answer

Lesions separated in time and space; internuclear ophthalmoplegia (impaired adduction with contralateral abducting nystagmus) is a classic finding.

#377 · Nervous System and Special Senses

Myasthenia gravis: mechanism, hallmark clinical pattern, and confirmatory antibody?

Answer

Autoantibodies against postsynaptic acetylcholine receptors; fatigable weakness worse with use and better with rest, starting with ptosis/diplopia; confirmed with acetylcholine receptor antibodies.

#378 · Nervous System and Special Senses

Guillain-Barre syndrome: typical trigger, pattern of weakness, and the most dangerous complication?

Answer

Follows an infection (classically *Campylobacter jejuni*) by 1 to 3 weeks; ascending, symmetric weakness with areflexia; the most dangerous complication is respiratory failure.

#379 · Nervous System and Special Senses

Most common cause of peripheral neuropathy in the outpatient setting, and its typical pattern?

Answer

Diabetes mellitus; symmetric, distal, sensory-predominant polyneuropathy in a stocking-glove distribution.

#380 · Nervous System and Special Senses

Bacterial meningitis: classic cerebrospinal fluid pattern and empiric adult treatment?

Answer

Neutrophil-predominant pleocytosis, low glucose, high protein; empiric treatment is vancomycin plus a third generation cephalosporin, plus ampicillin if *Listeria* risk factors are present.

#381 · Nervous System and Special Senses

Adjunctive medication in bacterial meningitis that reduces neurologic complications, and its timing?

Answer

Dexamethasone, given with or just before the first antibiotic dose.

#382 · Nervous System and Special Senses

Herpes simplex encephalitis: classic imaging finding and urgent treatment?

Answer

Temporal lobe edema/signal change on MRI; start empiric intravenous acyclovir immediately once suspected.

#383 · Nervous System and Special Senses

Acute angle closure glaucoma: classic exam findings and urgent treatment?

Answer

Fixed, mid-dilated pupil, firm globe, halos around lights, severe pain; treat urgently with pressure-lowering agents followed by laser peripheral iridotomy.

#384 · Nervous System and Special Senses

Retinal detachment: classic symptoms and urgency of referral?

Answer

Sudden floaters, flashes of light, and a curtain-like visual field defect; a surgical emergency requiring urgent ophthalmology referral.

#385 · Nervous System and Special Senses

Weber and Rinne test findings in conductive vs sensorineural hearing loss?

Answer

Conductive: Weber lateralizes to the affected ear, bone conduction better than air conduction on Rinne in that ear. Sensorineural: Weber lateralizes to the better ear, normal Rinne pattern (air better than bone) in both ears.

#386 · Respiratory System

What spirometry finding confirms reversible airflow obstruction in asthma?

Answer

An FEV1/FVC ratio that improves by at least 12 percent and 200 mL after bronchodilator administration.

#387 · Respiratory System

Why should a long-acting beta agonist never be used as monotherapy in asthma?

Answer

Without a concurrent inhaled corticosteroid it can mask worsening airway inflammation and increases the risk of severe exacerbation and death.

#388 · Respiratory System

In a tachypneic asthmatic, why is a rising or normalizing PaCO₂ an ominous sign rather than reassuring?

Answer

It indicates the patient can no longer sustain the hyperventilation that had been driving CO₂ down, signaling impending respiratory muscle fatigue and failure.

#389 · Respiratory System

What spirometry value defines COPD?

Answer

A postbronchodilator FEV1/FVC ratio below 0.70 that does not normalize with bronchodilator use.

#390 · Respiratory System

What is the only intervention proven to slow disease progression and reduce mortality in COPD?

Answer

Smoking cessation.

#391 · Respiratory System

What are the thresholds for prescribing long-term home oxygen in stable COPD?

Answer

Resting PaO₂ at or below 55 mmHg, or oxygen saturation at or below 88 percent, or PaO₂ at or below 59 mmHg with cor pulmonale or polycythemia.

#392 · Respiratory System

What is first-line respiratory support for a COPD exacerbation with respiratory acidosis in a patient who can protect the airway?

Answer

Noninvasive positive pressure ventilation (BiPAP), which reduces work of breathing and often avoids intubation.

#393 · Respiratory System

What condition should be suspected in a young or nonsmoking patient with lower lobe predominant panacinar emphysema?

Answer

Alpha-1 antitrypsin deficiency; check a serum alpha-1 antitrypsin level.

#394 · Respiratory System

What is the empiric outpatient antibiotic choice for a previously healthy patient with community-acquired pneumonia and no comorbidities?

Answer

Amoxicillin or a macrolide.

#395 · Respiratory System

What five criteria make up the CURB-65 score for pneumonia severity?

Answer

Confusion, urea (BUN) elevation, respiratory rate at or above 30, blood pressure below 90 systolic or 60 diastolic, and age 65 or older; a score of 2 or more supports admission.

#396 · Respiratory System

What clinical clues suggest Legionella pneumonia, and how is it diagnosed?

Answer

High fever with relative bradycardia, GI symptoms, confusion, and hyponatremia; diagnosed with a urinary antigen test.

#397 · Respiratory System

Why does Legionella pneumonia not respond to beta-lactam antibiotics?

Answer

Legionella is an intracellular organism, and beta-lactams, which target cell wall synthesis, do not achieve adequate intracellular concentrations; macrolides or fluoroquinolones are used instead.

#398 · Respiratory System

Why does aspiration pneumonia characteristically affect the right lower lobe?

Answer

The right mainstem bronchus is more vertical and more in line with the trachea than the left, directing aspirated material there preferentially.

#399 · Respiratory System

What organisms should be empirically covered in hospital-acquired or ventilator-associated pneumonia?

Answer

Pseudomonas aeruginosa and other resistant gram-negative rods, plus MRSA coverage (vancomycin or linezolid) when risk factors are present.

#400 · Respiratory System

What are Light's criteria for an exudative pleural effusion?

Answer

Pleural fluid to serum protein ratio above 0.5, pleural fluid to serum LDH ratio above 0.6, or pleural fluid LDH above two-thirds the upper limit of normal serum LDH; meeting any one criterion classifies the effusion as exudative.

#401 · Respiratory System

What pleural fluid findings in a parapneumonic effusion mandate chest tube drainage rather than antibiotics alone?

Answer

pH below 7.2, glucose below 60, a positive Gram stain, or frank pus (empyema).

#402 · Respiratory System

In a patient with low pretest probability for pulmonary embolism, what test can effectively exclude the diagnosis?

Answer

A negative D-dimer; in moderate-to-high pretest probability, proceed directly to CT pulmonary angiography instead.

#403 · Respiratory System

What is the imaging test of choice for suspected PE in pregnancy, and why?

Answer

Ventilation-perfusion (V/Q) scanning, because it delivers less radiation to radiosensitive breast tissue than CT pulmonary angiography.

#404 · Respiratory System

What treatment should be considered in addition to anticoagulation for a hemodynamically unstable (massive) PE?

Answer

Systemic thrombolysis or catheter-directed therapy to rapidly restore right ventricular function.

#405 · Respiratory System

When is an inferior vena cava filter indicated in venous thromboembolism?

Answer

A contraindication to anticoagulation, or recurrent PE despite adequate, confirmed therapeutic anticoagulation.

#406 · Respiratory System

What clinical findings define tension pneumothorax, and what is the immediate treatment?

Answer

Tracheal deviation away from the affected side, hypotension, distended neck veins, and absent breath sounds unilaterally; treat with immediate needle decompression at the second intercostal space, midclavicular line, before imaging.

#407 · Respiratory System

What paraneoplastic syndromes are classically associated with small cell lung cancer?

Answer

SIADH, ectopic ACTH secretion causing Cushing syndrome, and Lambert-Eaton myasthenic syndrome.

#408 · Respiratory System

How does Lambert-Eaton myasthenic syndrome differ clinically from myasthenia gravis?

Answer

Lambert-Eaton strength improves with repeated muscle use (presynaptic calcium channel antibodies), whereas myasthenia gravis worsens with repeated use (postsynaptic acetylcholine receptor antibodies).

#409 · Respiratory System

What mechanism causes hypercalcemia in squamous cell lung carcinoma?

Answer

Tumor secretion of parathyroid hormone-related protein (PTHrP), which mimics PTH action while suppressing endogenous PTH.

#410 · Respiratory System

What is a Pancoast tumor, and what syndrome does it cause?

Answer

An apical lung tumor invading the brachial plexus (shoulder/arm pain) and sympathetic chain, causing Horner syndrome: ptosis, miosis, and anhidrosis.

#411 · Respiratory System

What pulmonary nodule features suggest a benign process rather than malignancy?

Answer

Small size, smooth margins, and central, popcorn, or laminated calcification patterns.

#412 · Respiratory System

What pulmonary function pattern defines interstitial lung disease?

Answer

A restrictive pattern: reduced FEV1 and FVC with a preserved or increased FEV1/FVC ratio, and a reduced DLCO.

#413 · Respiratory System

What high resolution CT pattern and demographic are classic for idiopathic pulmonary fibrosis?

Answer

Older men with a smoking history; usual interstitial pneumonia pattern with peripheral, basal predominant reticular changes and honeycombing.

#414 · Respiratory System

What occupational exposure and radiographic pattern define silicosis, and what infection risk does it raise?

Answer

Mining, sandblasting, or quarry work; upper lobe predominant nodules with eggshell calcification of hilar nodes; increased susceptibility to tuberculosis.

#415 · Respiratory System

What occupational exposure and radiographic pattern define asbestosis, and what malignancies does it increase risk for?

Answer

Shipbuilding, insulation, or construction work; lower lobe predominant fibrosis with pleural plaques; increased risk of mesothelioma and, especially with concurrent smoking, bronchogenic carcinoma.

#416 · Respiratory System

What extrapulmonary findings and mechanism explain hypercalcemia in sarcoidosis?

Answer

Erythema nodosum, uveitis, and hypercalcemia from unregulated conversion of vitamin D to its active form by macrophages within noncaseating granulomas.

#417 · Respiratory System

Why does reactivation tuberculosis characteristically involve the upper lung zones?

Answer

Mycobacterium tuberculosis is an obligate aerobe, and the upper lobes have relatively higher oxygen tension and lower blood flow, favoring organism growth.

#418 · Respiratory System

What adverse effects are classically linked to isoniazid, rifampin, ethambutol, and pyrazinamide in TB treatment?

Answer

Isoniazid: peripheral neuropathy (prevented with pyridoxine) and hepatotoxicity. Rifampin: body fluid discoloration and CYP450 induction. Ethambutol: optic neuritis with red-green color blindness. Pyrazinamide: hepatotoxicity and hyperuricemia.

#419 · Respiratory System

What apnea-hypopnea index (AHI) thresholds define mild, moderate, and severe obstructive sleep apnea, and what is first-line treatment?

Answer

Mild 5-15, moderate 15-30, severe above 30 events per hour; continuous positive airway pressure (CPAP) is first-line treatment across severities.

#420 · Respiratory System

What are the Berlin criteria for ARDS, and what ventilator and positioning strategies reduce mortality?

Answer

Onset within one week of a known insult, bilateral opacities not fully explained by effusion/collapse/nodules, and respiratory failure not fully explained by cardiac failure or fluid overload; low tidal volume ventilation (about 6 mL/kg predicted body weight) and prone positioning in moderate-to-severe disease reduce mortality.

#421 • Multisystem Processes and Disorders

What are the three components of the qSOFA score used to screen for possible sepsis at the bedside?

Answer

Altered mentation, respiratory rate 22 or higher, and systolic blood pressure 100 mmHg or less. Two or more of these three findings in a patient with suspected infection should prompt closer monitoring.

#422 • Multisystem Processes and Disorders

How is septic shock defined, distinct from sepsis alone?

Answer

Sepsis with persistent hypotension requiring vasopressors to maintain a mean arterial pressure of 65 mmHg or higher, plus a serum lactate above 2 mmol/L, despite adequate fluid resuscitation.

#423 · Multisystem Processes and Disorders

What is the first-line vasopressor in septic shock, and why?

Answer

Norepinephrine, because it provides balanced alpha and beta adrenergic support with a lower risk of tachyarrhythmia than dopamine, which is no longer preferred as first-line therapy.

#424 · Multisystem Processes and Disorders

What hemodynamic profile distinguishes septic (distributive) shock from cardiogenic shock?

Answer

Septic shock shows a low systemic vascular resistance with a normal or elevated cardiac output, especially early on. Cardiogenic shock shows a low cardiac output with elevated filling pressures because the heart itself has failed as a pump.

#425 · Multisystem Processes and Disorders

What is the correct order of the ATLS primary survey?

Answer

Airway with cervical spine protection, Breathing, Circulation with hemorrhage control, Disability (rapid neurologic check), and Exposure with environmental control (ABCDE), ordered by what kills a patient fastest.

#426 · Multisystem Processes and Disorders

What clinical findings define a tension pneumothorax, and what is the immediate treatment?

Answer

Absent breath sounds on the affected side, tracheal deviation away from that side, distended neck veins, and hypotension. Treat immediately with needle decompression followed by tube thoracostomy, without waiting for imaging.

#427 · Multisystem Processes and Disorders

What are the three components of the lethal triad of trauma?

Answer

Hypothermia, acidosis, and coagulopathy. Each of these three worsens the other two, creating a vicious cycle that must be actively interrupted with warming and appropriate resuscitation.

#428 · Multisystem Processes and Disorders

What is the modern balanced transfusion strategy for massive hemorrhage in trauma?

Answer

Packed red blood cells, fresh frozen plasma, and platelets given in roughly equal ratios, rather than crystalloid-heavy resuscitation or red cells alone, to better replace lost components and reduce dilutional coagulopathy.

#429 · Multisystem Processes and Disorders

What are the physical findings of the cholinergic toxidrome, and what is the treatment?

Answer

Salivation, lacrimation, urination, defecation, gastrointestinal distress, emesis, plus miosis and bradycardia (organophosphate poisoning). Treat with atropine for muscarinic symptoms and pralidoxime to reactivate acetylcholinesterase before it becomes permanently bound.

#430 · Multisystem Processes and Disorders

How does the anticholinergic toxidrome differ from the sympathomimetic toxidrome, given that both cause tachycardia, hypertension, and mydriasis?

Answer

Anticholinergic toxicity produces dry, flushed skin, while sympathomimetic toxicity (cocaine, amphetamines) produces diaphoresis (sweaty skin). This skin finding is the key distinguishing feature between the two toxidromes.

#431 · Multisystem Processes and Disorders

What is the antidote for acetaminophen overdose, and how does it work?

Answer

N-acetylcysteine, which works by replenishing glutathione stores that would otherwise be depleted, allowing accumulation of the toxic metabolite NAPQI, which causes centrilobular hepatocellular necrosis.

#432 · Multisystem Processes and Disorders

What acid-base pattern is classic for early salicylate toxicity?

Answer

A mixed picture: an early respiratory alkalosis from direct stimulation of the medullary respiratory center, followed by an evolving anion gap metabolic acidosis as salicylate uncouples oxidative phosphorylation.

#433 · Multisystem Processes and Disorders

What is the antidote for methanol or ethylene glycol poisoning, and what does it do?

Answer

Fomepizole, which blocks alcohol dehydrogenase and prevents formation of the toxic metabolites responsible for the acidosis and organ injury. Hemodialysis is added for severe toxicity.

#434 · Multisystem Processes and Disorders

What is the Revised Cardiac Risk Index used for, and what are its six factors?

Answer

It estimates perioperative cardiac risk before noncardiac surgery. The six factors are high-risk surgery, ischemic heart disease, congestive heart failure, cerebrovascular disease, insulin-dependent diabetes, and a preoperative creatinine above 2 mg/dL.

#435 · Multisystem Processes and Disorders

How should an established beta blocker be managed perioperatively, and why should a new one not be started acutely?

Answer

Continue an existing beta blocker through surgery to avoid rebound tachycardia or ischemia. Do not start one acutely on the day of surgery, since new initiation has been linked to increased perioperative stroke and mortality.

#436 · Multisystem Processes and Disorders

What is the typical timing and differential for postoperative fever at each interval?

Answer

Days 1-2: inflammatory response and possible atelectasis. Days 3-5: urinary tract infection. Days 5-7: surgical site infection or anastomotic leak. After day 7: deep venous thrombosis or drug fever.

#437 · Multisystem Processes and Disorders

What findings suggest necrotizing soft tissue infection rather than a routine wound infection?

Answer

Rapidly spreading skin discoloration, bullae, crepitus, pain out of proportion to examination findings, and systemic toxicity. This is a surgical emergency requiring immediate operative debridement plus broad-spectrum antibiotics.

#438 · Multisystem Processes and Disorders

What serologic findings track disease activity in lupus nephritis flares, and why?

Answer

A rising anti-double-stranded DNA antibody titer with falling C3 and C4 complement levels, because complement is being consumed by immune complex deposition in the glomeruli during active disease.

#439 · Multisystem Processes and Disorders

Why should high-dose glucocorticoids be started immediately in suspected giant cell arteritis, without waiting for biopsy?

Answer

Untreated giant cell arteritis risks permanent, irreversible blindness. Temporal artery biopsy findings can remain positive for one to two weeks after starting steroids, so treatment should never be delayed for confirmation.

#440 · Multisystem Processes and Disorders

How do the three ANCA-associated small-vessel vasculitides differ from one another?

Answer

Granulomatosis with polyangiitis affects the upper and lower respiratory tract plus kidneys with c-ANCA. Microscopic polyangiitis is p-ANCA associated without granulomas. Eosinophilic granulomatosis with polyangiitis adds asthma and eosinophilia, also p-ANCA associated.

#441 • Multisystem Processes and Disorders

What renal complication of systemic sclerosis presents with abrupt severe hypertension and acute kidney injury?

Answer

Scleroderma renal crisis. Early treatment with an angiotensin-converting enzyme inhibitor significantly improves outcomes, so prompt recognition is critical.

#442 • Multisystem Processes and Disorders

What are the four traditional categories in the differential diagnosis of fever of unknown origin?

Answer

Infections (tuberculosis, endocarditis, occult abscess), malignancy (such as lymphoma or renal cell carcinoma), autoimmune or inflammatory conditions (such as adult-onset Still disease or vasculitis), and miscellaneous causes (such as drug fever).

#443 • Multisystem Processes and Disorders

What distinguishes Janeway lesions from Osler nodes in infective endocarditis?

Answer

Janeway lesions are painless erythematous macules on the palms and soles from septic microemboli. Osler nodes are painful, tender nodules on the finger and toe pads from immune complex deposition.

#444 • Multisystem Processes and Disorders

What is the appropriate immediate treatment for hyperkalemia with electrocardiographic changes, and what is given first?

Answer

Intravenous calcium is given first to stabilize the cardiac membrane and reduce arrhythmia risk, even though it does not lower serum potassium. Insulin with glucose and potassium-removal measures follow.

#445 · Multisystem Processes and Disorders

Why does correcting chronic severe hyponatremia too quickly cause harm, and what is the safe correction rate?

Answer

Rapid correction (generally more than 8 to 10 mEq/L in the first 24 hours) risks osmotic demyelination syndrome, a serious and often irreversible neurologic injury. Correction should proceed slowly with frequent monitoring.

#446 · Endocrine System

In DKA, why can the initial serum potassium be normal or high despite total body potassium depletion?

Answer

Insulin deficiency and acidosis shift potassium out of cells into the serum, masking true total body depletion. Check potassium before starting insulin; if below 3.3 mEq/L, replete potassium first, since insulin will drive it back into cells and can cause dangerous hypokalemia.

#447 · Endocrine System

How do DKA and HHS differ in their typical glucose, pH, and ketone findings?

Answer

DKA usually shows glucose above 250 mg/dL, pH below 7.3, and positive ketones, occurring mainly in type 1 diabetes. HHS shows much higher glucose, often above 600 mg/dL, near-normal pH, and absent or minimal ketones, occurring in type 2 diabetes with residual insulin secretion.

#448 · Endocrine System

What is the correct sequence for adding dextrose during a DKA insulin infusion?

Answer

Once glucose falls to about 200 to 250 mg/dL, add dextrose to the intravenous fluids while continuing the insulin infusion. This allows insulin to keep clearing ketosis without causing hypoglycemia; stopping insulin too early leaves ketogenesis unchecked.

#449 · Endocrine System

What diagnostic criteria establish type 2 diabetes mellitus?

Answer

Fasting plasma glucose of 126 mg/dL or higher, a 2-hour oral glucose tolerance test value of 200 mg/dL or higher, hemoglobin A1c of 6.5 percent or higher, or a random glucose of 200 mg/dL or higher with classic symptoms. Confirm on a separate day unless symptoms are unmistakable.

#450 · Endocrine System

Which diabetes medication classes have proven cardiovascular benefit beyond glucose lowering?

Answer

GLP-1 receptor agonists and SGLT2 inhibitors both reduce major adverse cardiovascular events independent of their glycemic effect. They are preferred additions to metformin in patients with established atherosclerotic disease, heart failure, or chronic kidney disease.

#451 · Endocrine System

What lab pattern distinguishes primary from secondary hypothyroidism?

Answer

Primary hypothyroidism (thyroid gland failure) shows a high TSH with low free T4. Secondary hypothyroidism (pituitary or hypothalamic failure) shows a low or inappropriately normal TSH with low free T4, since the pituitary cannot mount the expected TSH response.

#452 · Endocrine System

What antibody and gland finding support a diagnosis of Hashimoto thyroiditis?

Answer

Anti-thyroid peroxidase and anti-thyroglobulin antibodies are typically positive, and the thyroid gland is firm, nontender, and diffusely enlarged on exam. It is the most common cause of hypothyroidism in iodine-sufficient countries like the United States.

#453 · Endocrine System

Why must fever or sore throat prompt urgent evaluation in a patient taking methimazole?

Answer

Methimazole carries a risk of agranulocytosis, and fever or sore throat can be the only early warning sign. The medication should be held and a complete blood count checked promptly rather than assuming a routine viral illness.

#454 · Endocrine System

Which antithyroid drug is preferred in the first trimester of pregnancy, and why?

Answer

Propylthiouracil is preferred in the first trimester because methimazole carries a rare risk of embryopathy during this period. A switch back to methimazole is often made after the first trimester to reduce propylthiouracil's own hepatotoxicity risk.

#455 · Endocrine System

What radioactive iodine uptake pattern distinguishes Graves disease from subacute thyroiditis?

Answer

Graves disease shows high, diffuse radioactive iodine uptake from new hormone synthesis. Subacute thyroiditis shows low uptake because the thyrotoxic phase comes from inflammatory release of preformed hormone, not new production.

#456 · Endocrine System

What is the correct medication sequence for treating thyroid storm?

Answer

Give a beta blocker and an antithyroid drug such as propylthiouracil first, then give iodine solution at least one hour later. Giving iodine too early can provide substrate for new hormone synthesis before the antithyroid drug takes effect.

#457 · Endocrine System

What ultrasound features of a thyroid nodule raise suspicion for malignancy?

Answer

Microcalcifications, irregular margins, marked hypoechogenicity, and a height-to-width ratio greater than 1 all raise suspicion. These features, when TSH is normal or high, should prompt fine needle aspiration biopsy.

#458 · Endocrine System

What is the initial screening test for suspected Cushing syndrome, and why not a random cortisol?

Answer

A 24-hour urinary free cortisol, late-night salivary cortisol, or overnight low-dose dexamethasone suppression test. Cortisol follows a diurnal rhythm, so a single random morning value is unreliable for screening.

#459 · Endocrine System

Why does ectopic ACTH secretion from a lung tumor often cause more severe hypokalemia than pituitary Cushing disease?

Answer

Very high ACTH levels drive marked cortisol overproduction, and cortisol at high enough levels has mineralocorticoid-like effects on the kidney, promoting potassium wasting more prominently than typical pituitary-driven Cushing disease.

#460 · Endocrine System

Why is aldosterone typically preserved in secondary but not primary adrenal insufficiency?

Answer

Aldosterone secretion is primarily controlled by the renin-angiotensin system, not ACTH. Primary adrenal insufficiency destroys the whole cortex, losing both cortisol and aldosterone, while secondary disease from pituitary failure spares aldosterone production.

#461 · Endocrine System

Why must alpha blockade always precede beta blockade before pheochromocytoma surgery?

Answer

Starting a beta blocker first leaves alpha-adrenergic vasoconstriction unopposed, which can trigger a severe hypertensive crisis. Alpha blockade (typically phenoxybenzamine) must be established first, with beta blockade added afterward if needed.

#462 · Endocrine System

In panhypopituitarism, why must glucocorticoids be replaced before levothyroxine?

Answer

Levothyroxine accelerates cortisol metabolism. Starting it before glucocorticoid replacement can unmask or worsen underlying adrenal insufficiency and precipitate an adrenal crisis, so cortisol replacement always comes first.

#463 · Endocrine System

What laboratory pattern distinguishes primary from secondary hyperparathyroidism?

Answer

Primary hyperparathyroidism shows high calcium with high or inappropriately normal PTH, usually from a solitary adenoma. Secondary hyperparathyroidism shows low or low-normal calcium with high PTH, driven by chronic kidney disease or vitamin D deficiency.

#464 · Endocrine System

Why is PTH appropriately suppressed in hypercalcemia of malignancy from PTH-related peptide?

Answer

PTH-related peptide activates the same receptor as PTH on bone and kidney but is a distinct molecule not detected by a standard intact PTH assay. The elevated calcium properly suppresses true PTH secretion through normal feedback.

#465 · Endocrine System

What five criteria define metabolic syndrome, and how many must be met?

Answer

At least three of: abdominal obesity, triglycerides 150 mg/dL or higher, HDL below 40 in men or 50 in women, blood pressure 130/85 mmHg or higher, and fasting glucose 100 mg/dL or higher. It reflects underlying insulin resistance.

#466 · Pregnancy Childbirth and the Puerperium

What blood pressure and proteinuria values define preeclampsia?

Answer

New-onset blood pressure of 140/90 mm Hg or higher after 20 weeks gestation, plus either proteinuria (300 mg or more per 24-hour urine, protein-to-creatinine ratio of 0.3 or greater, or 2+ on dipstick) or evidence of end-organ dysfunction if proteinuria is absent.

#467 · Pregnancy Childbirth and the Puerperium

What laboratory and blood pressure findings define severe features of preeclampsia?

Answer

Systolic blood pressure 160 mm Hg or higher or diastolic 110 mm Hg or higher on two readings four hours apart, platelet count below 100,000/microliter, creatinine above 1.1 mg/dL or doubled, liver transaminases at least twice normal, pulmonary edema, or new headache or visual disturbance.

#468 · Pregnancy Childbirth and the Puerperium

What does the acronym HELLP stand for and what is it a variant of?

Answer

Hemolysis, Elevated Liver enzymes, and Low Platelets; it is a severe variant of preeclampsia that can occur with or without frank hypertension and carries risk of hepatic rupture and disseminated intravascular coagulation.

#469 · Pregnancy Childbirth and the Puerperium

What is the first sign of magnesium sulfate toxicity, and what is the antidote?

Answer

Loss of deep tendon reflexes is the earliest sign, occurring before respiratory depression and cardiac arrest at higher levels. The antidote is calcium gluconate.

#470 · Pregnancy Childbirth and the Puerperium

Which antihypertensive classes are contraindicated throughout pregnancy, and why?

Answer

ACE inhibitors and angiotensin receptor blockers are contraindicated throughout pregnancy because of fetal renal toxicity; they may be used postpartum.

#471 · Pregnancy Childbirth and the Puerperium

Who should receive low-dose aspirin for preeclampsia prevention, and when should it start?

Answer

Patients with one high-risk factor (prior preeclampsia, multifetal gestation, chronic hypertension, diabetes, renal disease, or autoimmune disease) or two or more moderate risk factors, starting at 12 to 28 weeks and ideally before 16 weeks.

#472 · Pregnancy Childbirth and the Puerperium

What are the two approaches to gestational diabetes screening and their diagnostic thresholds?

Answer

Two-step: 50g one-hour challenge, if abnormal confirm with 100g three-hour OGTT (two or more abnormal values diagnostic). One-step: 75g two-hour OGTT, one abnormal value (fasting 92, one-hour 180, two-hour 153 mg/dL) is diagnostic.

#473 · Pregnancy Childbirth and the Puerperium

Why does gestational diabetes not increase the risk of congenital anomalies, unlike pregestational diabetes?

Answer

Gestational diabetes typically develops in the late second trimester, after organogenesis is already complete, so hyperglycemia does not affect fetal organ formation the way poorly controlled pregestational diabetes does around conception.

#474 · Pregnancy Childbirth and the Puerperium

What is the discriminatory zone in early pregnancy, and how is it used?

Answer

The beta-hCG level (roughly 1,500 to 2,000 mIU/mL) above which a gestational sac should be visible on transvaginal ultrasound in a normal intrauterine pregnancy; if no sac is seen above this level, ectopic pregnancy must be suspected.

#475 · Pregnancy Childbirth and the Puerperium

What is the expected rise in beta-hCG in a normal early intrauterine pregnancy?

Answer

Beta-hCG roughly doubles every 48 to 72 hours in a normally progressing early intrauterine pregnancy; a slower rise, plateau, or fall suggests a failing pregnancy or an ectopic pregnancy.

#476 · Pregnancy Childbirth and the Puerperium

What criteria must be met for methotrexate treatment of an unruptured ectopic pregnancy?

Answer

Hemodynamic stability, no fetal cardiac activity, hCG generally below about 5,000 mIU/mL, mass size under about 3.5 to 4 cm, and no contraindication such as breastfeeding, immunodeficiency, hepatic or renal disease, or peptic ulcer disease.

#477 · Pregnancy Childbirth and the Puerperium

How do threatened, inevitable, incomplete, complete, and missed abortion differ?

Answer

Threatened: closed os, viable pregnancy, with bleeding. Inevitable: open os, no tissue expelled yet. Incomplete: open os, some tissue retained. Complete: empty uterus after full expulsion. Missed: fetal demise with no expulsion and a closed os.

#478 · Pregnancy Childbirth and the Puerperium

What is the standard workup for recurrent pregnancy loss?

Answer

Testing for antiphospholipid syndrome, evaluation of uterine anatomy, parental karyotype for chromosomal translocations, and screening for uncontrolled thyroid disease or diabetes.

#479 · Pregnancy Childbirth and the Puerperium

What are the three tiers of fetal heart rate tracing interpretation, and how is each managed?

Answer

Category I (normal baseline, moderate variability, no late or variable decelerations) is reassuring. Category II is indeterminate, managed with observation. Category III (absent variability with recurrent late or variable decelerations, bradycardia, or sinusoidal pattern) requires immediate intrauterine resuscitation and expedited delivery if it does not improve.

#480 · Pregnancy Childbirth and the Puerperium

What distinguishes early, late, and variable decelerations by mechanism?

Answer

Early decelerations mirror the contraction and reflect fetal head compression (benign). Late decelerations begin after the contraction peak and reflect uteroplacental insufficiency (concerning). Variable decelerations have abrupt, variably timed onset and reflect umbilical cord compression.

#481 · Pregnancy Childbirth and the Puerperium

How is an arrest disorder in the active phase of labor defined and managed?

Answer

No cervical change for four hours with adequate contractions (or six hours with inadequate contractions) after 6 cm dilation with ruptured membranes; first-line management is oxytocin augmentation or amniotomy, with cephalopelvic disproportion diagnosed only after adequate contractions fail to produce progress.

#482 · Pregnancy Childbirth and the Puerperium

What is the first-line management for shoulder dystocia, and what maneuver should never be used?

Answer

The McRoberts maneuver (hyperflexing the maternal hips) combined with suprapubic pressure is first-line. Fundal pressure should never be used, since it worsens impaction of the shoulder.

#483 · Pregnancy Childbirth and the Puerperium

What are the four Ts of postpartum hemorrhage, and which is most common?

Answer

Tone (uterine atony), Tissue (retained placenta), Trauma (lacerations or rupture), and Thrombin (coagulopathy). Uterine atony (Tone) accounts for about 70 to 80% of cases.

#484 · Pregnancy Childbirth and the Puerperium

How does postpartum psychosis differ from the baby blues and postpartum depression in timing and severity?

Answer

The baby blues begin within days and resolve within about two weeks with mild mood lability and no impairment. Postpartum depression persists beyond two weeks with functional impairment. Postpartum psychosis presents within the first two weeks with hallucinations, delusions, or thoughts of harming the infant, and is a psychiatric emergency requiring immediate hospitalization.

#485 · Pregnancy Childbirth and the Puerperium

When is anti-D immune globulin (Rhogam) given, and when is it not useful?

Answer

Given at 28 weeks, within 72 hours of delivery if the newborn is Rh-positive, and after any sensitizing event, to prevent maternal antibody formation before sensitization occurs. It has no benefit once a patient is already alloimmunized.

#486 · Biostatistics Epidemiology and Population Health

What is the formula for sensitivity, and what does a highly sensitive test rule out?

Answer

Sensitivity = true positives / (true positives + false negatives). A highly sensitive test has few false negatives, so a negative result helps rule the disease out (SnNout).

#487 · Biostatistics Epidemiology and Population Health

What is the formula for specificity, and what does a highly specific test rule in?

Answer

Specificity = true negatives / (true negatives + false positives). A highly specific test has few false positives, so a positive result helps rule the disease in (SpPin).

#488 · Biostatistics Epidemiology and Population Health

What is the formula for positive predictive value (PPV), and how does prevalence affect it?

Answer

PPV = true positives / (true positives + false positives). As disease prevalence falls, PPV falls, since a larger fraction of positive results become false positives.

#489 · Biostatistics Epidemiology and Population Health

What is the formula for negative predictive value (NPV), and how does prevalence affect it?

Answer

$NPV = \text{true negatives} / (\text{true negatives} + \text{false negatives})$. As disease prevalence falls, NPV rises, since true negatives make up a larger share of the population.

#490 · Biostatistics Epidemiology and Population Health

How are the positive and negative likelihood ratios calculated from sensitivity and specificity?

Answer

$LR+ = \text{sensitivity} / (1 - \text{specificity})$. $LR- = (1 - \text{sensitivity}) / \text{specificity}$. Unlike PPV and NPV, likelihood ratios do not change with disease prevalence.

#491 · Biostatistics Epidemiology and Population Health

What is the formula for relative risk (RR), and which study design calculates it directly?

Answer

RR = incidence in the exposed group / incidence in the unexposed group. Cohort studies and randomized controlled trials calculate RR directly, since both measure incidence in each group.

#492 · Biostatistics Epidemiology and Population Health

What is the formula for the odds ratio (OR), and which study design uses it?

Answer

OR = $(a \times d) / (b \times c)$ from a 2x2 table of exposed/unexposed cases and controls. Case-control studies use OR because they cannot directly calculate incidence.

#493 · Biostatistics Epidemiology and Population Health

What is the rare disease assumption, and why does it matter?

Answer

When a disease is rare in the source population, the odds ratio from a case-control study closely approximates the relative risk a cohort study would have found.

#494 · Biostatistics Epidemiology and Population Health

How is the number needed to treat (NNT) calculated, and what does a lower NNT mean?

Answer

$$\text{NNT} = 1 / \text{absolute risk reduction (control event rate minus experimental event rate)}$$

A lower NNT means fewer patients need treatment for one additional patient to benefit.

#495 · Biostatistics Epidemiology and Population Health

How is the number needed to harm (NNH) calculated, and what does a higher NNH mean?

Answer

$NNH = 1 / \text{absolute risk increase (treatment event rate minus control event rate)}$. A higher NNH means more patients must be exposed before one is harmed, which is more favorable.

#496 · Biostatistics Epidemiology and Population Health

What is the difference between incidence and prevalence?

Answer

Incidence is new cases over a period divided by the population at risk. Prevalence is all existing cases at one point in time divided by the total population. Prevalence approximately equals incidence times average disease duration.

#497 · Biostatistics Epidemiology and Population Health

What is lead-time bias, and how does it distort apparent survival after screening?

Answer

Lead-time bias occurs when early detection lengthens the known duration of disease without changing the actual time of death, making screen-detected patients appear to survive longer even though nothing about the disease course changed.

#498 · Biostatistics Epidemiology and Population Health

What is length-time bias, and why does it favor screen-detected disease?

Answer

Length-time bias occurs because screening preferentially detects slower-growing, less aggressive disease that stays detectable longer, while fast-growing disease is more likely to present with symptoms between screenings, making the screened group look healthier than the truth.

#499 · Biostatistics Epidemiology and Population Health

What is the difference between a confounder and an effect modifier?

Answer

A confounder is a third variable independently linked to both exposure and outcome that distorts the true association and should be adjusted for. An effect modifier genuinely changes the strength of an association across strata and should be reported separately, not adjusted away.

#500 · Biostatistics Epidemiology and Population Health

What do a type I error, a type II error, and statistical power each represent?

Answer

Type I error (α) is a false positive: rejecting a true null hypothesis. Type II error (β) is a false negative: failing to reject a false null hypothesis. Power ($1 - \beta$) is the probability of correctly detecting a true effect, and it increases with larger sample size and effect size.